Supplementary Diagnostics for a Reproducible CR3BP Low-Thrust Cislunar Benchmark Resource

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Purpose

This supplement preserves secondary diagnostics, implementation-heavy tables, and small-seed evidence for the normalized Earth–Moon CR3BP missed-thrust benchmark resource. The main article reports the compact claim path; the full audit tables are retained here so provenance and caveats remain inspectable without making the article a full artifact listing. The claim-ledger artifacts are `data/results/claim_evidence_ledger/claim_evidence_ledger.csv`, `data/results/claim_evidence_ledger/claim_evidence_ledger_metadata.json`, `data/results/claim_evidence_ledger/tail_coast_threshold_audit.csv`, and `data/results/claim_evidence_ledger/tail_coast_branch_audit.csv`; they replay recorded artifacts only and do not rerun optimization. The independent-midpoint Hermite–Simpson all-configured headroom artifacts are `data/results/independent_hs_all_configured_headroom/independent_hs_all_configured_headroom.csv`, `data/results/independent_hs_all_configured_headroom/independent_hs_all_configured_headroom_metadata.json`, `tables/independent_hs_all_configured_headroom/independent_hs_all_configured_headroom_table.tex`, and `figures/`

independent_hs_all_configured_headroom/independent_hs_all_configured_headroom.pdf; they are normalized-CR3BP all-configured one-segment evidence only. The replay/stress artifacts in data/results/replay_stress_validation/ now include the all-configured independent-midpoint Hermite–Simpson row's endpoint and midpoint nominal sidecar; they check nominal-control replay, integration-grid refinement, and small acceleration scaling only. The independent-HS branch-control replay artifacts are data/results/independent_hs_branch_control_replay/independent_hs_branch_control_replay.csv, data/results/independent_hs_branch_control_replay/independent_hs_branch_control_replay_metadata.json, and tables/independent_hs_branch_control_replay/independent_hs_branch_control_replay_table.tex; they validate sidecar hashes and repropagate persisted endpoint-plus-midpoint branch controls under normalized CR3BP only. The independent-HS bicircular phase-stress artifacts are data/results/independent_hs_bicircular_phase_stress/independent_hs_bicircular_phase_stress.csv, data/results/independent_hs_bicircular_phase_stress/independent_hs_bicircular_phase_stress_metadata.json, and tables/independent_hs_bicircular_phase_stress/independent_hs_bicircular_phase_stress_table.tex; they replay persisted endpoint-plus-midpoint controls under a simple circular solar-tidal stress model and are not SPICE or high-fidelity validation. The independent-HS cached-Horizons-derived solar-tidal replay artifacts are data/cache/horizons/independent_hs_phase_shift_2026jan01_vectors.json, data/results/independent_hs_horizons_solar_tidal_replay/independent_hs_horizons_solar_tidal_replay.csv, data/results/independent_hs_horizons_solar_tidal_replay/independent_hs_horizons_solar_tidal_replay_metadata.json, and tables/independent_hs_horizons_solar_tidal_replay/independent_hs_horizons_solar_tidal_replay_table.tex; they replay persisted controls with representative 2026-Jan-01 cached JPL Horizons geometry and remain a simplified stress probe, not SPICE/high-fidelity validation or broad production-solver validation/parity. The independent-HS cached-Horizons Earth/Moon/Sun point-mass retuning artifacts are data/results/independent_hs_horizons_point_mass_retuning/independent_hs_horizons_point_mass_retuning.csv, data/results/independent_hs_horizons_point_mass_retuning/independent_hs_horizons_point_mass_retuning_metadata.json, data/results/independent_hs_horizons_point_mass_retuning/controls/independent_hs_horizons_point_mass_retuned_control_manifest.json, and tables/independent_hs_horizons_point_mass_retuning/independent_hs_horizons_point_mass_retuning_table.tex; they document failed direct point-mass replay of the persisted controls and then retune independently at the representative 2026-Jan-01 cached-Horizons epoch, but they are not SPICE/full high-fidelity/flight validation, broad production-solver validation/parity, fuel optimality, or quantum evidence. The independent-HS multi-epoch cached-Horizons point-mass retuning artifacts are data/results/independent_hs_horizons_multi_epoch_point_mass_retuning/independent_hs_horizons_multi_epoch_point_mass_retuning.csv, data/results/independent_hs_horizons_multi_epoch_point_mass_retuning/independent_hs_horizons_multi_epoch_point_mass_retuning_metadata.json, data/results/independent_hs_horizons_multi_epoch_point_mass_retuning/epochs/, and tables/independent_hs_horizons_multi_epoch_point_mass_retuning/independent_hs_horizons_multi_epoch_point_mass_retuning_table.tex; they repeat that stress/retuning protocol at four fixed 2026 cached-Horizons epochs and remain point-mass retuning evidence only, not SPICE/full high-fidelity/flight validation, broad production-solver validation/parity, fuel optimality, DOI evidence, or quantum evidence. The independent-HS SPICE-derived ephemeris replay artifacts are the compact caches data/cache/spice/independent_hs_phase_shift_2026*_spice_vectors.json, data/results/independent_hs_spice_ephemeris_replay/independent_hs_spice_ephemeris_replay.csv, data/results/independent_hs_ `spice_ephemeris_replay/independent_hs_spice_ephemeris_replay_metadata.json`, and `tables/independent_hs_spice_ephemeris_replay/independent_hs_spice_ephemeris_replay_table.tex`; they replay the 36 already-retuned controls under SPICE-derived Moon/Sun geocentric vectors and remain point-mass ephemeris-source replay, not full high-fidelity/flight validation, broad production-solver validation/parity, fuel optimality, DOI evidence, or quantum evidence. The focused tail-coast branch-control replay artifacts are `data/results/hard_catalog_tail_coast_branch_control_replay/tail_coast_branch_control_replay.csv`, `data/results/hard_catalog_tail_coast_branch_control_replay/tail_coast_branch_control_replay_metadata.json`, and `tables/hard_catalog_tail_coast_branch_control_replay/tail_coast_branch_control_replay_table.tex`; they re-propagate persisted accepted controls under normalized CR3BP only. The Horizons contrast artifacts are `data/cache/horizons/hard_catalog_tail_coast_2026jan01_vectors.json`, `data/results/horizons_ephemeris_force_model_contrast/horizons_ephemeris_force_model_contrast.csv`, `data/results/horizons_ephemeris_force_model_contrast/horizons_ephemeris_force_model_contrast_metadata.json`, and `tables/horizons_ephemeris_force_model_contrast/horizons_ephemeris_force_model_contrast_table.tex`; they compare cached Earth/Moon/Sun geometry and solar-tidal accelerations only and are not SPICE or high-fidelity validation. The hard-catalog bicircular stress artifacts are `data/results/bicircular_solar_tidal_stress/bicircular_solar_tidal_stress.csv`, `data/results/bicircular_solar_tidal_stress/bicircular_solar_tidal_stress_metadata.json`, and `tables/bicircular_solar_tidal_stress/bicircular_solar_tidal_stress_table.tex`; they are a deterministic beyond-CR3BP stress replay and not high-fidelity validation. The bicircular retuned recovery artifacts are `data/results/bicircular_tail_coast_recovery/bicircular_tail_coast_recovery.csv`, `data/results/bicircular_tail_coast_recovery/bicircular_tail_coast_recovery_summary.csv`, `data/results/bicircular_tail_coast_recovery/bicircular_tail_coast_recovery_metadata.json`, and `tables/bicircular_tail_coast_recovery/bicircular_tail_coast_recovery_table.tex`; they retune under a simple fixed-phase bicircular model but remain a negative threshold result. The synthesis artifacts are `data/results/evidence_synthesis/evidence_synthesis.csv`, `data/results/evidence_synthesis/evidence_synthesis_metadata.json`, and `tables/evidence_synthesis/evidence_synthesis_table.tex`; they replay recorded artifacts only and do not rerun optimization.

1 Claim-Ledger and Replay Audit Tables

The tables in this section are generated audit artifacts. They distinguish evidence semantics and replay recorded controls; they do not rerun trajectory optimization, add high-fidelity validation, or certify operational robustness.

Table 1: Reviewer-facing claim evidence ledger generated from recorded artifacts only.

Claim row	Evidence family	Target	Mask scope	All configured?	Interpretation	Boundary
phase_shift_main_method_30seed_selected_branch	30-seed phase-shift main-method initializer comparison	halo phase-shift	30 seeds; selected one-segment branch-recovery benchmark	False	No sampled-method or QAOA superiority; all-windows continuous has lower paired selected-worst error than every sampled method in this benchmark.	Selected-branch statistical evidence only; no quantum advantage and no all-configured-mask claim.
phase_shift_qaoa_qubo_30seed_selected_branch	30-seed QAOA/QUBO ablation	halo phase-shift	30 seeds; QUBO/QAOA sampler-family ablation on selected one-segment benchmark	False	Optimized p=2 QAOA is competitive with surrogate-QUBO SA in this ablation, but paired tests do not support statistical superiority (mean selected-worst delta=-0.000558832, sign-test p=0.454498).	Simulated QAOA ablation only; no hardware, quantum-advantage, or superiority claim.
phase_shift_continuation_all_single_p04_all_configured	continuation extension	halo phase-shift	8/8 configured one-segment masks selected and evaluated	True	All configured one-segment phase-shift masks pass in this continuation row.	One-segment N=8 phase-shift scope only; not a QUBO/QAOA or high-fidelity claim.
phase_shift_continuation_two_segment_n8_p03_all_configured	continuation extension	halo phase-shift	15/15 configured one- and two-segment masks selected and evaluated	True	All configured N=8 one- and two-segment phase-shift masks pass in this continuation row.	N=8 one/two-mask phase-shift scope only; not broader outage-family robustness.
phase_shift_direct_collocation_p04_selected_branch_diagnostic	direct-collocation baseline	halo phase-shift	one selected branch optimized; all configured one-segment masks evaluated diagnostically	False	The selected p=0.4 branch is feasible and all-mask diagnostic is below configured thresholds.	Selected-branch continuous-backend diagnostic; all masks were not selected for optimization.
phase_shift_independent_hs_p04_amax02_selected_branch_diagnostic	independent-midpoint Hermite-Simpson baseline	halo phase-shift	3 selected branches optimized; 8 one-segment masks evaluated diagnostically	False	Independent midpoint controls give a strong phase-shift continuous-backend diagnostic.	Selected-branch/all-mask distinction is preserved; not all-configured-mask evidence.
phase_shift_independent_hs_p04_amax02_all_configured	independent-midpoint Hermite-Simpson all-configured headroom	halo phase-shift	8/8 configured one-segment masks selected and evaluated	True	The new independent-midpoint HS package gives all-configured one-segment phase-shift headroom at p=0.4 and amax=0.2 in normalized CR3BP.	Normalized-CR3BP all-configured one-segment evidence only; optimizer reached the max_nfev cap, and this is not broad production-solver validation/parity, high-fidelity validation, fuel optimality, broader outage-family robustness, QUBO, QAOA, or quantum evidence.
phase_shift_independent_hs_branch_control_replay	independent-midpoint Hermite-Simpson branch-control replay	halo phase-shift	16 branch replay rows across 2 all-configured one-segment independent-HS rows; 8 branches per row	True	Persisted independent-HS branch controls replay deterministically under normalized CR3BP with max terminal-error delta 0.0; the polish row converges (optimizer_success=True, nfev=25) but has slightly higher selected/all error than the capped source row.	Branch-control replay only; no optimization rerun, high-fidelity validation, broad production-solver validation/parity, fuel optimality, broader outage-family robustness, quantum, QUBO, or QAOA claim.
phase_shift_independent_hs_bicircular_phase_stress_probe	independent-HS simple bicircular phase-sweep stress replay	halo phase-shift	converged polish row; 8 nominal phase rows and 64 branch-phase rows over Sun phases 0.0, 45.0, 90.0, 135.0, 180.0, 225.0, 270.0, 315.0	True	Positive simple bicircular phase-sweep stress replay for the converged independent-HS all-configured row; persisted controls remain within configured thresholds across the tested phases.	Deterministic beyond-CR3BP stress probe only; not SPICE ephemeris validation, high-fidelity flight validation, broad production-solver validation/parity, fuel optimality, quantum, QUBO, or QAOA evidence.
phase_shift_independent_hs_horizons_solar_tidal_replay_probe	independent-HS cached-Horizons-derived solar-tidal replay	halo phase-shift	representative 2026-Jan-01 epoch; polish row has 1 nominal replay and 8 configured one-segment branch replays	True	This is stronger than the simple bicircular phase sweep because the Sun geometry is derived from a cached JPL Horizons Moon/Sun vector cache; the persisted polish-row controls remain within the configured thresholds for this representative stress replay.	Cached-Horizons-derived solar-tidal stress probe only; not SPICE validation, high-fidelity or flight validation, accepted-control high-fidelity replay, broad production-solver validation/parity, fuel optimality, quantum, QUBO, or QAOA evidence.
phase_shift_independent_hs_horizons_point_mass_retuning	independent-HS cached-Horizons Earth/Moon/Sun point-mass retuning	halo phase-shift	representative 2026-Jan-01 epoch; persisted nominal plus 8 branch controls directly replayed and independently retuned	True	Persisted controls fail direct ephemeris point-mass replay, but independent retuning restores feasibility for nominal and all 8 branches at the representative epoch.	Cached-Horizons Earth/Moon/Sun point-mass retuning stress model only; not SPICE/full high-fidelity/flight validation, not broad production-solver validation/parity, not fuel optimality, not DOI evidence, and not quantum, QUBO, or QAOA evidence.
phase_shift_independent_hs_horizons_multi_epoch_point_mass_retuning	independent-HS multi-epoch cached-Horizons Earth/Moon/Sun point-mass retuning	halo phase-shift	4 representative 2026 epochs (2026-Jan-01, 2026-Apr-01, 2026-Jul-01, 2026-Oct-01); persisted nominal plus 32 branch controls directly replayed and independently retuned	True	The single-epoch point-mass retuning concern is stress-tested across four cached-Horizons 2026 epochs: nominal direct replay fails in every epoch, while direct branch replay passes 18/32 overall (July 8/8) and independent retuning restores feasibility for nominal and all branch rows.	Multi-epoch cached-Horizons Earth/Moon/Sun point-mass retuning stress model only; not SPICE/full high-fidelity/flight validation, not broad production-solver validation/parity, not fuel optimality, not DOI evidence, and not quantum, QUBO, or QAOA evidence.
phase_shift_independent_hs_spice_ephemeris_replay	independent-HS SPICE-derived ephemeris replay	halo phase-shift	4 fixed 2026 epochs (2026-Jan-01, 2026-Apr-01, 2026-Jul-01, 2026-Oct-01); 4 nominal and 32 already-retuned branch controls replayed without retuning	True	The existing multi-epoch point-mass retuned controls replay under SPICE-derived geometric Moon/Sun vectors with branch pass count 32/32 and max error delta 3.2763991519857427e-10 relative to the Horizons-retuned replay.	SPICE-derived ephemeris replay under the paper Earth/Moon/Sun point-mass model only; no retuning or optimization rerun, not full high-fidelity/flight validation, not production solver parity, not fuel optimality, not DOI evidence, and not quantum, QUBO, or QAOA evidence.
phase_shift_independent_hs_casadi_ipopt_bridge	independent-HS scoped CasADi/IPOPT mature NLP backend bridge check	halo phase-shift	1 nominal plus 8 configured one-segment branch rows from the accepted polish case	True	One normalized-CR3BP accepted independent-HS polish case can be exported to a mature CasADi/IPOPT NLP backend and locally replayed/refined under the same target, scales, branch masks, and source branch-recovery semantics.	Scoped CasADi/IPOPT mature NLP backend bridge check only for one normalized-CR3BP accepted case; local direct-shooting refinement, not broad production-solver validation/parity, not production mission design, not high-fidelity or flight validation, global optimality, fuel optimality, DOI evidence, not quantum, QUBO, or QAOA evidence.
catalog_dro_tail_coast_all_one_two_segment_t5_all_configured	hard-catalog tail-coast fixed-final-time recovery	catalog-DRO	27/27 configured one- and two-segment masks selected and evaluated	True	The combined hard-catalog tail-coast row passes configured fixed-final-time thresholds for all configured one- and two-segment masks.	Fixed-final-time tail-coast locked-nominal backend only; no high-fidelity, fuel-optimal, quantum, QUBO, or QAOA claim.
catalog_dro_tail_coast_branch_control_replay_accepted_controls	hard-catalog tail-coast accepted branch-control replay	catalog-DRO	27/27 persisted accepted branch-control sidecars replayed	True	Persisted accepted controls replay deterministically under the normalized CR3BP model with max terminal-error delta 0.0 across nominal plus 27 branch rows.	Accepted-control replay only; no optimization rerun, high-fidelity validation, broad production-solver validation/parity, fuel optimality, quantum, QUBO, or QAOA claim.
catalog_dro_tail_coast_bicircular_solar_tidal_stress_probe	bicircular solar-tidal stress replay	catalog-DRO	108 branch-phase stress rows from 27 accepted branch sidecars and Sun phases 0.0, 90.0, 180.0, 270.0	False	The persisted controls reproduce exactly in CR3BP but do not remain threshold-feasible under the configured bicircular solar-tidal stress sweep.	Beyond-CR3BP stress probe only; not SPICE ephemeris validation, high-fidelity flight validation, broad production-solver validation/parity, fuel optimality, quantum, QUBO, or QAOA evidence.
bicircular_tail_coast_retuned_recovery	bicircular tail-coast retuned recovery, negative	catalog-DRO	fixed Sun phase 0.0 deg; all 27/27 configured one- and two-segment masks retuned; package_complete=True	True	Completed fixed-phase simple bicircular retuning lowers the nominal error from 2.481764912283746 to 0.31677192167859453 but still fails: nominal configured pass=False, branch pass count=19/27, max branch error=6.029904532225566, strict branch pass count=16/27.	Simple bicircular solar-tidal retuning at fixed phase 0.0 deg only; original fixed target and final time; not SPICE/high-fidelity/flight validation, broad production-solver validation/parity, fuel optimality, quantum, QUBO, or QAOA evidence.
catalog_dro_tail_coast_horizons_ephemeris_force_model_contrast	Horizons ephemeris force-model contrast	catalog-DRO	15 hard-catalog transfer nodes; accepted nominal controls and representative branch mask 22 sampled for acceleration contrast	False	Cached Horizons geometry quantifies Earth-Moon distance/rate variation and fixed-reference Sun distance/phase differences missing from the simple circular model over the hard-catalog transfer nodes.	Ephemeris force-model contrast only; not SPICE validation, high-fidelity flight validation, accepted-control high-fidelity replay, broad production-solver validation/parity, fuel optimality, quantum, QUBO, or QAOA evidence.
catalog_dro_delayed_h6_all_single_delayed_arrival	hard-catalog delayed-arrival recovery	catalog-DRO	14/14 configured one-segment masks selected and evaluated against the delayed target	True	The h6 portfolio recovers all one-segment masks against the delayed-arrival target.	Delayed-arrival evidence only; not fixed-final-time robustness, not fuel optimality, and not quantum/QUBO/QAOA evidence.

Table 2: Tail-coast recorded-error threshold audit for `tail_coast_all_one_two_segment_t5_portfolio`.

Threshold pair	Nominal threshold	Selected/all threshold	Nominal pass	Selected pass	All-mask pass	Pair pass
configured_0p09_0p17	0.09	0.17	True	True	True	True
screen_0p05_0p12	0.05	0.12	True	True	True	True
near_margin_0p025_0p095	0.025	0.095	True	True	True	True
robust_tight_0p025_0p09	0.025	0.09	True	False	False	False
very_tight_0p02_0p09	0.02	0.09	False	False	False	False

Table 3: Tail-coast branch audit parsed from the historical `branch_results` JSON field.

Case	Branches	Optimizer ran	Optimizer success	Direct no-recovery	Fallback accepted	Max terminal error	Accepted weights	Accepted init kinds
tail_coast_all_one_two_segment_t5_portfolio	27	25	25	2	4	0.0936063931709301	regularized_001=6; terminal_only=21	constant_vector=4; no_recovery_variables=2; nominal_post_outage=21

Table 4: Deterministic nominal-control replay/stress validation from recorded sidecars, including endpoint-plus-midpoint replay for the independent-midpoint Hermite–Simpson rows. This table checks nominal sidecar replay only and does not rerun missed-thrust branch recovery.

Case	Role/backend	Replay variant	Nominal error	Delta from recorded	Configured nominal pass	Tighter nominal pass	Tight nominal pass	Interpretation/limits
all_single_p04_warm_from_p03	bounded multiple-shooting continuous continuation; endpoint nominal controls	baseline_source_substeps	0.0530981	7.63278e-17	True	True	False	Deterministic source-grid nominal-control replay; branch recovery controls are not replayed.
all_single_p04_warm_from_p03	bounded multiple-shooting continuous continuation; endpoint nominal controls	refined_2x_substeps	0.0530981	-3.20058e-09	True	True	False	Integration-grid refinement only; target definition and recorded controls are fixed.
all_single_p04_warm_from_p03	bounded multiple-shooting continuous continuation; endpoint nominal controls	refined_4x_substeps	0.0530981	-3.40584e-09	True	True	False	Integration-grid refinement only; target definition and recorded controls are fixed.
all_single_p04_warm_from_p03	bounded multiple-shooting continuous continuation; endpoint nominal controls	accel_scale_0p99_source_substeps	0.0537741	0.000676002	True	True	False	Small direct acceleration-scale stress around recorded controls; not high-fidelity validation.
all_single_p04_warm_from_p03	bounded multiple-shooting continuous continuation; endpoint nominal controls	accel_scale_1p01_source_substeps	0.052471	-0.000627051	True	True	False	Small direct acceleration-scale stress around recorded controls; may exceed the configured acceleration bound and is not high-fidelity validation.
two_segment_n8_p03_cold	bounded multiple-shooting continuous continuation; endpoint nominal controls	baseline_source_substeps	0.0612012	2.77556e-17	True	True	False	Deterministic source-grid nominal-control replay; branch recovery controls are not replayed.
two_segment_n8_p03_cold	bounded multiple-shooting continuous continuation; endpoint nominal controls	refined_2x_substeps	0.0612012	8.95398e-10	True	True	False	Integration-grid refinement only; target definition and recorded controls are fixed.
two_segment_n8_p03_cold	bounded multiple-shooting continuous continuation; endpoint nominal controls	refined_4x_substeps	0.0612012	9.51416e-10	True	True	False	Integration-grid refinement only; target definition and recorded controls are fixed.
two_segment_n8_p03_cold	bounded multiple-shooting continuous continuation; endpoint nominal controls	accel_scale_0p99_source_substeps	0.057364	-0.0038372	True	True	False	Small direct acceleration-scale stress around recorded controls; not high-fidelity validation.
two_segment_n8_p03_cold	bounded multiple-shooting continuous continuation; endpoint nominal controls	accel_scale_1p01_source_substeps	0.0650786	0.00387734	True	False	False	Small direct acceleration-scale stress around recorded controls; may exceed the configured acceleration bound and is not high-fidelity validation.
ihs_phase_p04_amax02_warm_from_p03	independent-midpoint Hermite-Simpson collocation; endpoint and midpoint nominal controls	baseline_source_substeps	0.0197148	2.42861e-17	True	True	True	Deterministic source-grid nominal-control replay; branch recovery controls are not replayed.
ihs_phase_p04_amax02_warm_from_p03	independent-midpoint Hermite-Simpson collocation; endpoint and midpoint nominal controls	refined_2x_substeps	0.0197148	1.38038e-08	True	True	True	Integration-grid refinement only; target definition and recorded controls are fixed.
ihs_phase_p04_amax02_warm_from_p03	independent-midpoint Hermite-Simpson collocation; endpoint and midpoint nominal controls	refined_4x_substeps	0.0197148	1.46595e-08	True	True	True	Integration-grid refinement only; target definition and recorded controls are fixed.
ihs_phase_p04_amax02_warm_from_p03	independent-midpoint Hermite-Simpson collocation; endpoint and midpoint nominal controls	accel_scale_0p99_source_substeps	0.0201157	0.000400939	True	True	True	Small direct acceleration-scale stress around recorded controls; not high-fidelity validation.
ihs_phase_p04_amax02_warm_from_p03	independent-midpoint Hermite-Simpson collocation; endpoint and midpoint nominal controls	accel_scale_1p01_source_substeps	0.0194994	-0.000215353	True	True	True	Small direct acceleration-scale stress around recorded controls; may exceed the configured acceleration bound and is not high-fidelity validation.
ihs_all_single_p04_amax02_warm_from_p03	all-configured independent-midpoint Hermite-Simpson; endpoint and midpoint nominal controls	baseline_source_substeps	0.0111152	0	True	True	True	Deterministic source-grid nominal-control replay; branch recovery controls are not replayed.
ihs_all_single_p04_amax02_warm_from_p03	all-configured independent-midpoint Hermite-Simpson; endpoint and midpoint nominal controls	refined_2x_substeps	0.0111152	-1.64567e-08	True	True	True	Integration-grid refinement only; target definition and recorded controls are fixed.
ihs_all_single_p04_amax02_warm_from_p03	all-configured independent-midpoint Hermite-Simpson; endpoint and midpoint nominal controls	refined_4x_substeps	0.0111152	-1.74763e-08	True	True	True	Integration-grid refinement only; target definition and recorded controls are fixed.
ihs_all_single_p04_amax02_warm_from_p03	all-configured independent-midpoint Hermite-Simpson; endpoint and midpoint nominal controls	accel_scale_0p99_source_substeps	0.0108719	-0.000243327	True	True	True	Small direct acceleration-scale stress around recorded controls; not high-fidelity validation.
ihs_all_single_p04_amax02_warm_from_p03	all-configured independent-midpoint Hermite-Simpson; endpoint and midpoint nominal controls	accel_scale_1p01_source_substeps	0.0117006	0.000585421	True	True	True	Small direct acceleration-scale stress around recorded controls; may exceed the configured acceleration bound and is not high-fidelity validation.

Table 5: Deterministic independent-HS branch-control replay from persisted endpoint-plus-midpoint sidecars. The replay table is generated from recorded manifests and sidecars only and is limited to normalized CR3BP propagation.

Case	Branch rows	Nominal delta	Max branch delta	Max delta	Selected-worst delta	All-mask-worst delta	Tolerance	Pass
ihs_all_single_p04_amax02_warm_from_p03	8	0.000e+00	0.000e+00	0.000e+00	0.000e+00	0.000e+00	1.000e-10	True
ihs_all_single_p04_amax02_polish_from_p04	8	0.000e+00	0.000e+00	0.000e+00	0.000e+00	0.000e+00	1.000e-10	True

2 Catalog-Derived Schedule and QUBO Diagnostics

The original catalog-derived pilot remains negative evidence. QUBO simulated annealing and QAOA can produce competitive pre-refinement schedules in the small pilot, but no method meets the configured post-refinement robust thresholds after continuous recovery. The QUBO fit diagnostics show why surrogate validation is a first-order artifact rather than an assumption: with a full quadratic model and a small training set, training fit is much stronger than validation behavior.

Table 6: Catalog-derived pilot results. Values are medians across the generated pilot seeds. This benchmark did not meet the robust refinement thresholds.

Method	Refined nominal error	Selected recovery worst error	All-mask diagnostic worst error	Active fraction	Solver true evals	Shared QUBO train evals	Total true evals	Refine success
random	1.8309	2.5333	34.0442	0.3571	120.0000	0.0000	120.0000	0.0000
cross_entropy	0.9023	0.9031	9.9415	0.5000	120.0000	0.0000	120.0000	0.0000
genetic	2.5371	2.6132	34.8813	0.5000	132.0000	0.0000	132.0000	0.0000
true_sa	2.3540	2.4855	34.2025	0.6429	120.0000	0.0000	120.0000	0.0000
surrogate_qubo_sa	2.6497	2.7770	33.8472	0.5714	48.0000	72.0000	120.0000	0.0000
qaoa_statevector	2.4262	2.5573	33.8785	0.6429	48.0000	72.0000	120.0000	0.0000
all_windows_continuous	1.6383	2.2746	23.7862	1.0000	31.0000	0.0000	31.0000	0.0000

Table 7: Catalog-derived QUBO surrogate fit diagnostics.

seed	ridge	train_rmse	val_rmse	train_r2	val_r2	n_train	n_val	shared_qubo_training_evaluations	qubo_training_true_evaluations	equal_total_budget
11	0.0001	0.0002	4.3189	1.0000	0.5063	54	18	72	72	True
23	0.0001	0.0002	4.0060	1.0000	0.7190	54	18	72	72	True
37	0.0001	0.0002	5.9296	1.0000	0.0352	54	18	72	72	True

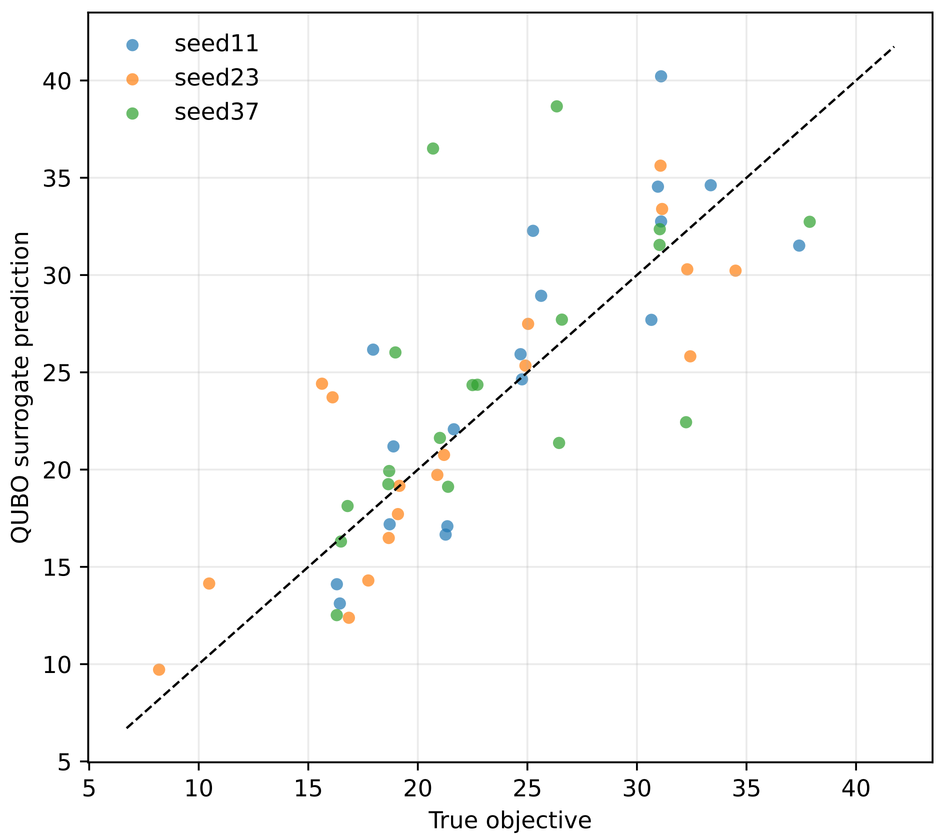

Figure 1: Catalog-derived QUBO fit diagnostic. The fit is reported as a diagnostic, not as evidence that the learned quadratic surrogate preserves the true trajectory ranking.

3 Catalog Feasibility and Multiple-Shooting Diagnostics

The feasibility sweep, direct multiple-shooting rows, targeted catalog attempt, and selected-outage envelope explain why the hard catalog-derived transfer is not promoted as a solved trajectory-design case. Selected recovery errors can be made small for chosen masks, but nominal errors, optimizer/backend success, and all-mask diagnostics remain limiting.

Table 8: Feasibility sweep stress test for the catalog-derived benchmark. None of the completed cases met both nominal and selected robust thresholds.

Transfer time	amax	Segments	Max nfev	Nominal error	Selected recovery worst error	All-outage diagnostic worst error	Meets thresholds	Best start	nfev	Runtime (s)
4.0000	0.2000	14	120	0.2008	0.2310	2.8975	False	feedback	251	1635.4279
4.0000	0.3000	14	150	0.1229	0.2783	7.5962	False	feedback	177	1102.4643
4.0000	0.3000	18	150	0.2423	0.2823	5.0379	False	feedback	39	351.0886
5.0000	0.3000	14	150	0.5810	0.6327	57.6216	False	zero	30	88.5080
5.0000	0.1500	14	120	0.6248	0.6785	53.6764	False	zero	138	997.2723
4.0000	0.2500	14	150	1.1558	1.2770	60.9139	False	feedback	81	324.7229
4.0000	0.1500	14	120	0.7389	1.6931	37.8087	False	zero	32	122.3204
3.0000	0.0650	14	120	1.0202	2.5174	7.1874	False	zero	61	326.9638

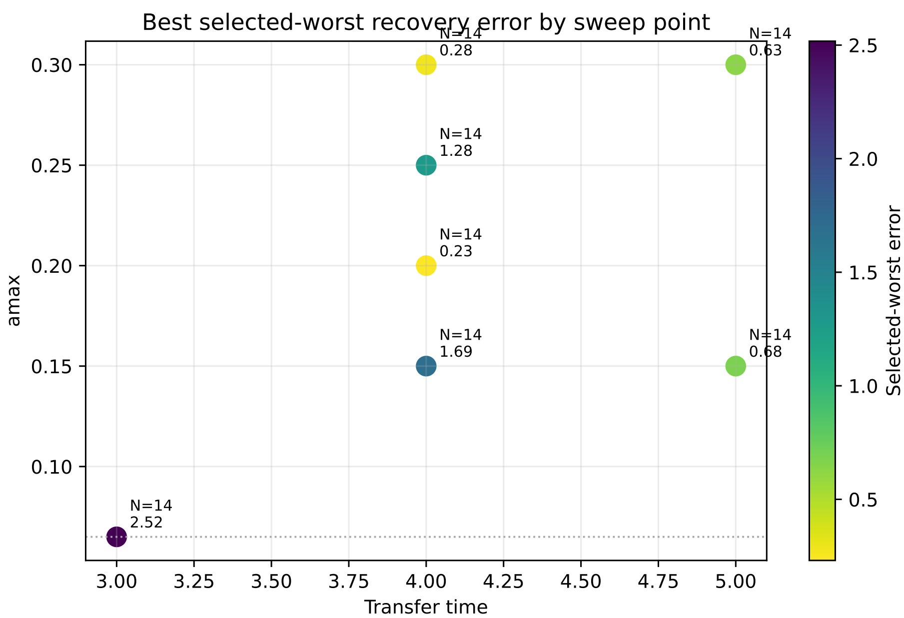

Figure 2: Feasibility sweep heatmap for the catalog-derived benchmark. The completed cases remain outside the configured robust success thresholds.

Table 9: Catalog-derived direct multiple-shooting diagnostics. Rows with zero selected outages are nominal-only diagnostics; all-outage diagnostic errors are still reported.

Transfer time	amax	Segments	Selected outages	Initialization	Nominal error	Selected recovery worst error	All-outage diagnostic worst error	Fuel	Max ——u——	Bound violation	nfev	Runtime (s)	Meets thresholds
4.0000	1.0000	14	1	blend	0.1714	0.0004	8.1180	4.0000	1.0000	0.0000	220	507.1263	False
5.0000	0.5000	14	0	linear	0.5154	0.5154	55.2845	2.5000	0.5000	0.0000	100	43.4353	False
4.0000	0.5000	14	3	linear	1.5355	1.2268	25.1426	2.0000	0.5000	0.0000	100	118.9605	False
4.0000	0.3000	14	3	linear	1.5981	1.9548	20.1304	1.2000	0.3000	0.0000	14	22.9601	False
4.0000	0.3000	14	3	linear	1.6273	4.3964	5.5503	1.2000	0.3000	0.0000	74	172.0316	False
5.0000	0.5000	14	3	linear	6.1753	5.7643	9.0650	2.5000	0.5000	0.0000	30	68.2684	False
6.0000	0.5000	14	0	linear	7.4830	7.4830	119.2412	3.0000	0.5000	0.0000	100	49.9072	False

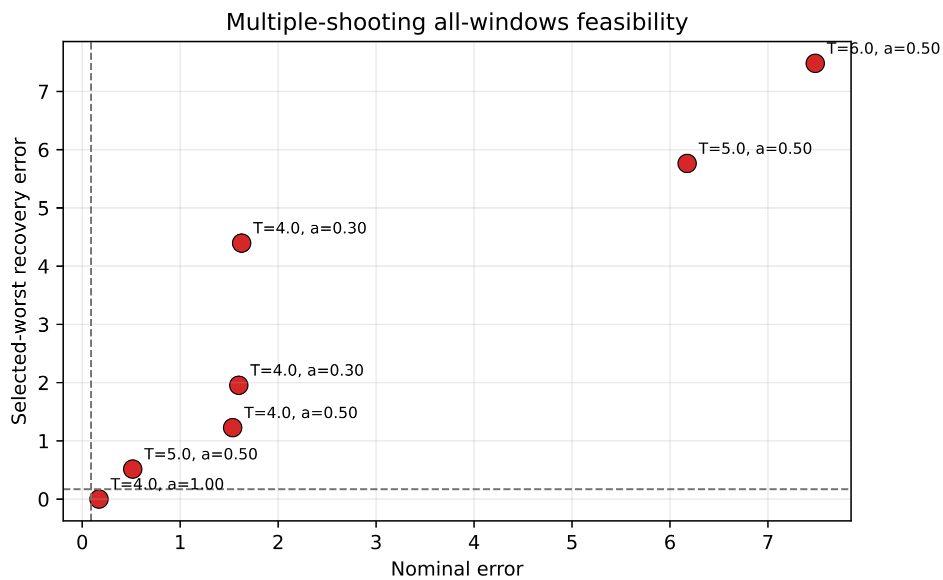

Figure 3: Catalog-derived multiple-shooting feasibility diagnostics. Selected-branch improvement does not remove the nominal and all-mask failure modes.

Table 10: Targeted catalog-derived feasibility attempt with stronger multistart branch recovery. Thresholds were not relaxed.

Transfer time	amax	Segments	Max nfev	Nominal error	Selected recovery worst error	All-outage diagnostic worst error	Meets thresholds	Best start	nfev	Runtime (s)
4.0000	0.3000	14	250	0.3055	1.3513	5.4449	False	zero	92	389.4414

Table 11: Selected-outage hard-catalog feasibility-envelope diagnostics. Selected recovery branches are optimized for the listed masks, but all rows fail the nominal threshold and all-mask values are diagnostics, not robustness claims.

Case	Method	Transfer time	amax	Segments	Selected outages	Nominal error	Selected worst error	All-mask diagnostic worst error	Max ——u——	Bound violation	nfev	Runtime (s)	Meets thresholds
selected_branch	multiple_shooting	4.0000	0.7000	14	1	0.3535	0.0360	21.8255	0.7000	0.0000	180	165.8788	False
selected_branch	multiple_shooting	4.0000	1.0000	14	1	0.8020	0.0009	9.3644	1.0000	0.0000	180	259.9800	False
selected_branch	multiple_shooting	4.0000	1.0000	14	3	1.6580	0.0150	8.7872	1.0000	0.0000	180	610.7961	False

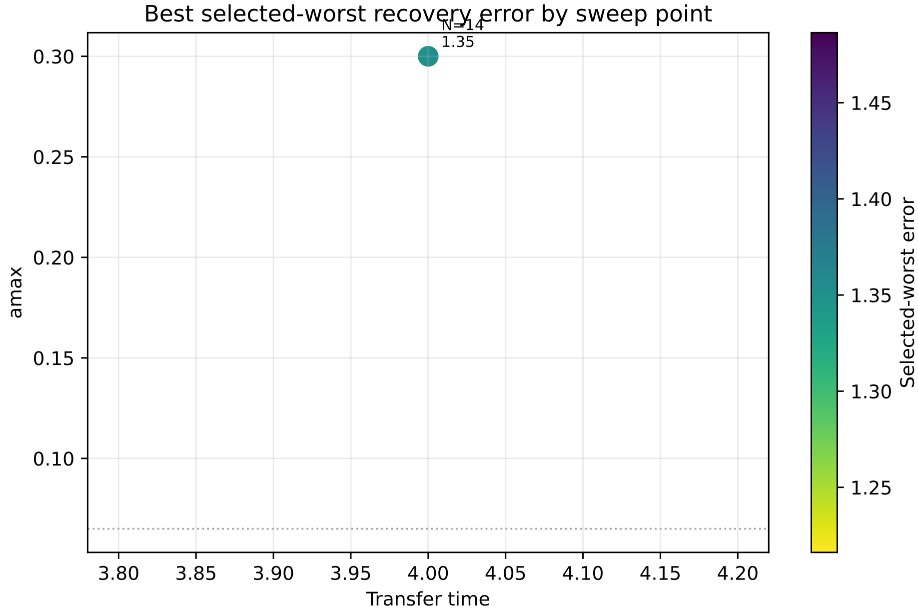

Figure 4: Targeted catalog-derived feasibility attempt. The stronger multistart attempt remains outside the configured thresholds.

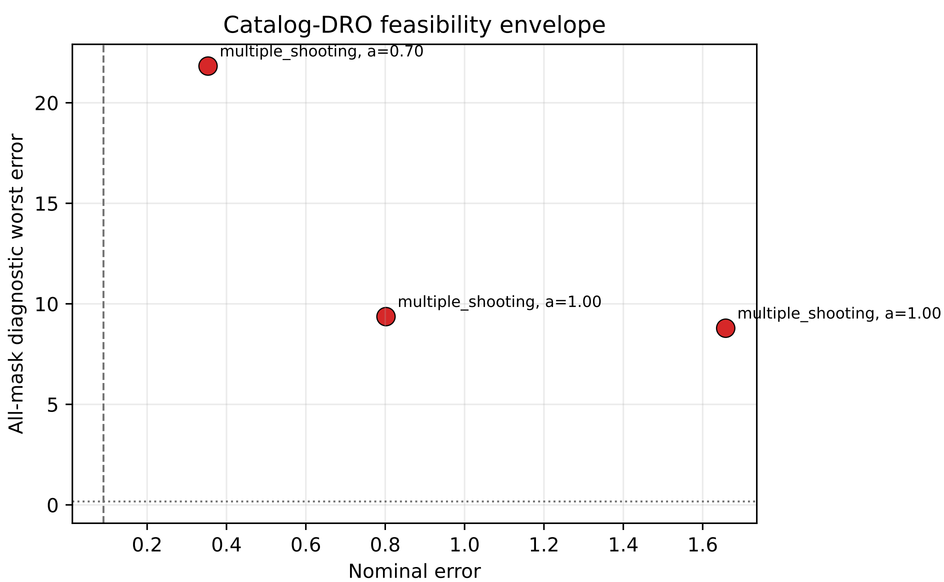

Figure 5: Selected-outage hard-catalog envelope. Small selected-outage errors do not imply robust success because nominal errors fail and all-mask diagnostics remain high.

4 Hard-Catalog Recovery Diagnostics

The locked-nominal, delayed-arrival, and tail-coast packages are continuous-backend diagnostics. They are not QUBO, QAOA, quantum, or discrete-sampler evidence. They are retained here to show the progression from selected-mask overfitting, to locked-nominal branch recovery, to delayed-arrival feasibility, and finally to the scoped fixed-final-time tail-coast row used in the main article.

Table 12: Locked-nominal hard-catalog branch-recovery diagnostics on the NRHO-like to DRO-phase benchmark. The nominal control is solved once and frozen; each selected missed-thrust branch is optimized independently after the outage. The all-mask column is a diagnostic over every configured mask, not a robustness claim. The `locked_hard_single_min6_selected8` row meets the thresholds but is not optimizer-converged.

Case	Policy	Selected masks	Nominal error	Selected worst error	All-mask diagnostic worst	Max ——u——	Bound violation	nfev	Runtime (s)	Meets thresholds
locked_hard_all_single	all_single	14	0.0131	0.8044	8.5410	1.0000	0.0000	955	575.7630	False
locked_hard_nominal_only	0	0	0.0131	0.0131	8.5410	1.0000	0.0000	71	40.6265	True
locked_hard_selected1	1	1	0.0131	0.1973	8.1384	1.0000	0.0000	161	96.3449	False
locked_hard_selected3	3	3	0.0131	0.1973	6.2972	1.0000	0.0000	401	247.4372	False
locked_hard_single_min6_selected8	8	8	0.0131	0.1580	1.4982	1.0000	0.0000	823	495.8345	True

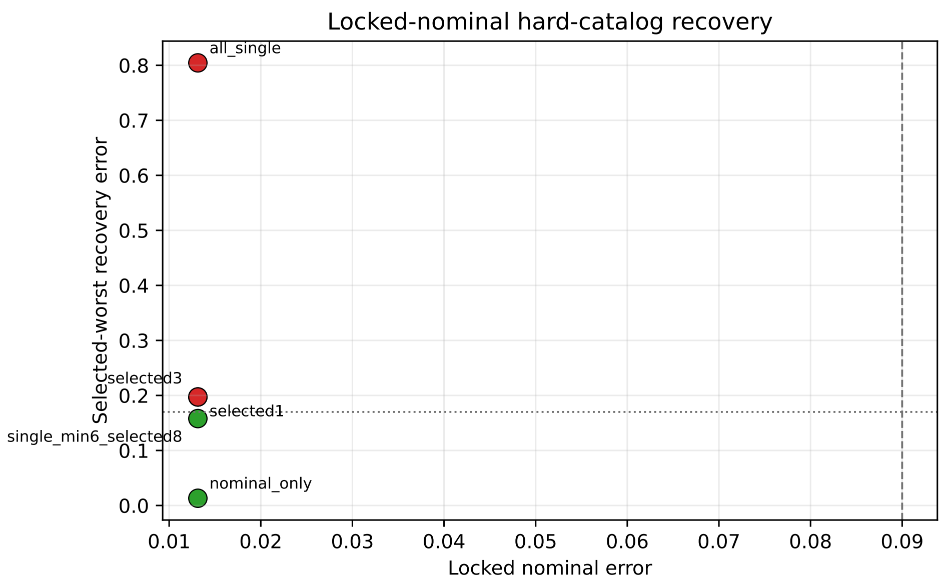

Figure 6: Locked-nominal hard-catalog branch recovery. With the nominal control held feasible, a bounded one-segment, six-recovery-segment selected subset meets the thresholds, while selected one/two-segment and all-single-outage scopes remain above the robust threshold.

The focused accepted branch-control replay package is present for the same combined tail-coast row. It reads the completed recovery CSV, nominal sidecar, manifest, and 27 persisted accepted branch-control sidecars, then repropagates those full-control schedules under the configured normalized CR3BP model. The generated table records zero nominal, maximum branch, and maximum terminal-error deltas at tolerance 10^{-10}, with `passes_tolerance=true`. This is a deterministic accepted-control replay only; it does not rerun optimization, branch portfolio selection, fallback search, high-fidelity validation, broad production-solver validation/parity checks, fuel-optimal analysis, or any quantum/QUBO/QAOA workflow.

The hard-catalog bicircular solar-tidal stress package reuses the same nominal and 27 accepted

Table 13: Delayed-arrival locked-nominal hard-catalog branch-recovery diagnostics. Branch duration is $(N+h)\Delta t$ and the branch target is the original target propagated ballistically by $h\Delta t$. The h6 portfolio row optimizes all 14 one-segment masks, charges all residual variants, and is delayed-arrival feasibility evidence only.

Case	Policy	Selected masks	Horizon segments	Portfolio	Variants	Nominal original error	Nominal delayed coast error	Selected delayed worst error	All-mask delayed diagnostic worst	Terminal weight	Control weight	Smooth weight	Continuity weight	Max ——u——	Bound violation	nfev	Runtime (s)	Meets thresholds	Nominal optimizer success	Branch optimizer ran	Branch optimizer all success
delayed_hard_all_single_h4_regularized	all_single	14	4	False	["configured"]	0.0131	0.0255	1.3990	1.3990	4.0000	0.0100	0.0100	0.0000	1.0000	0.0000	1338	1076.4599	False	True	True	False
delayed_hard_all_single_h6_portfolio	all_single	14	6	True	["regularized_001", "terminal_only"]	0.0131	0.0380	0.0040	0.0040	4.0000	0.0100	0.0100	0.0000	1.0000	0.0000	2253	2639.7199	True	True	True	True
delayed_hard_nominal_only_h4	0	0	4	False	["configured"]	0.0131	0.0255	0.0255	7.3102	4.0000	0.0100	0.0100	0.0000	1.0000	0.0000	71	37.4396	True	True	False	False

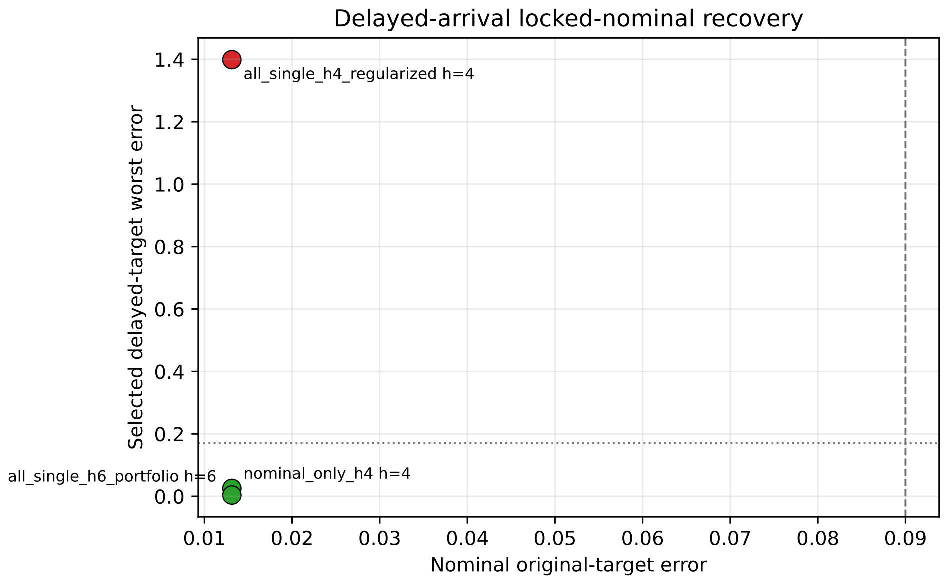

Figure 7: Delayed-arrival locked-nominal hard-catalog recovery. The h4 regularized all-single row fails, while the h6 branch-residual portfolio recovers all one-segment masks against the delayed target.

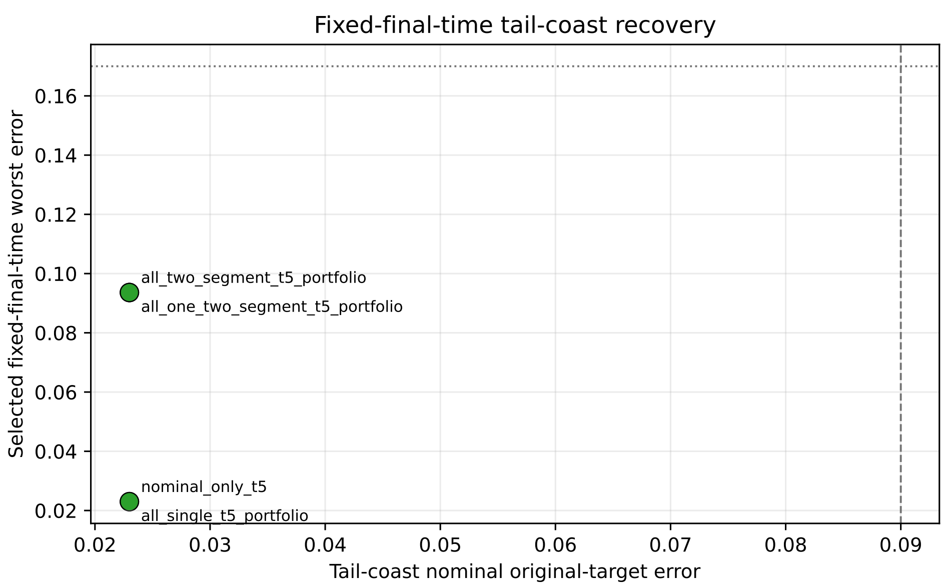

Figure 8: Tail-coast fixed-final-time hard-catalog recovery. The main article reports the table; this visual diagnostic is retained for the recovery-error pattern.

Table 14: Focused tail-coast accepted branch-control replay from persisted sidecars. The replay table is generated from recorded artifacts only and is limited to normalized CR3BP accepted-control determinism.

Case	Branch rows	Nominal delta	Max branch delta	Max delta	Tolerance	Pass
tail_coast_all_one_two_segment_t5_portfolio	27	0.000e+00	0.000e+00	0.000e+00	1.000e-10	True

branch-control sidecars. It adds a circular solar third-body tidal term with Sun distance 389.17 Earth–Moon distance units, $\mu_{\odot}/(\mu_E + \mu_M) = 328900.56$, Sun inertial angular-rate ratio 0.0748, and rotating-frame phase rate -0.9252. Unlike the positive independent-HS phase-shift stress replay below, this four-phase hard-catalog sweep is negative: CR3BP replay delta remains 0.0, but nominal solar-tidal rows fail the 0.09 threshold and only 22/108 branch-phase rows pass the 0.17 branch threshold. This is a stress-probe limitation, not SPICE ephemeris validation, high-fidelity flight validation, broad production-solver validation/parity, or fuel-optimal evidence.

Table 15: Bicircular solar-tidal stress replay from persisted tail-coast accepted-control sidecars. Controls are not retuned under the perturbation; failures are external-validity limitations of the persisted CR3BP controls.

Sun phase (deg)	Nominal solar-tidal error	Max branch solar-tidal error	Max branch delta	Branch passes	Branch rows	All branch pass
0	2.48176	5.40839	5.40804	5	27	False
90	0.877196	7.63031	7.6303	6	27	False
180	2.47851	5.3963	5.39596	5	27	False
270	0.897216	37.7596	37.7596	6	27	False

The bicircular retuned recovery batch then retunes the nominal and all 27 one/two-segment branch controls under the same simple bicircular solar-tidal model at fixed Sun phase 0°. It keeps the original fixed target and final time, so it is directly comparable to the fixed-final-time hard-catalog row. The batch is complete but negative: the initial bicircular nominal error 2.481764912283746 falls to a retuned nominal error of 0.316772, which still fails the configured 0.09 nominal threshold; configured branch pass count is 19/27, maximum retuned branch error is 6.0299, and the strict $(0.05, 0.09)$ branch pass count is 16/27. This is simple bicircular solar-tidal retuning only, not SPICE/high-fidelity/flight validation, broad production-solver validation/parity, fuel optimality, quantum, QUBO, or QAOA evidence.

Table 16: Bicircular tail-coast retuned recovery summary. The package retunes under a fixed-phase simple bicircular solar-tidal model but still fails the configured and strict threshold checks.

Sun phase (deg)	Retuned nominal error	Branch passes	Branch rows	Max retuned branch error	Pass 0.09/0.17	Pass 0.05/0.09	nfev
0	0.316772	19	27	6.0299	False	False	1027

The Horizons ephemeris force-model contrast package reads the committed cache in `data/cache/horizons/hard_catalog_tail_coast_2026jan01_vectors.json`. The cache stores exact JPL Horizons API URLs, query parameters, raw vector responses, and parsed Moon/Sun geocentric samples for the 15 hard-catalog segment nodes from a fixed representative 2026-Jan-01 TDB epoch. The generated contrast compares Earth–Moon distance variation, Earth–Moon angular-rate variation, fixed-reference Sun distance and rotating-frame phase variation, and solar-tidal acceleration differences at persisted nominal nodes plus the accepted branch with the largest recorded CR3BP terminal error. It is offline by default and is a force-model contrast only; it does not run SPICE, retune controls, or repropagate the accepted controls under ephemeris dynamics.

Table 17: Cached Horizons ephemeris force-model contrast from the focused tail-coast accepted-control package. The table reports model-assumption differences only and is not a validation pass/fail table.

Metric	Horizons-derived range	Simple-model reference	Interpretation
Earth–Moon distance	0.931638–1.04763	1.0 fixed	diagnostic ratio to window mean; not the Sun-vector LU scale
Earth–Moon angular rate	0.903116–1.1483	1.0 fixed	angular-rate variation absent from CR3BP normalization
Sun distance	382.686–382.896	389.17 LU; 384400 km/LU	cached Sun vector divided by the fixed CR3BP length unit
Sun phase offset	-7.88048–3.5743	-0.9252 rad/TU	aligned at first node, then compared to circular phase drift
Nominal tidal acceleration delta	max 0.00219803	bicircular solar tide	force contrast at persisted nominal trajectory nodes
Representative branch tidal acceleration delta	max 0.00219803	bicircular solar tide	force contrast at the accepted branch with largest recorded CR3BP terminal error

5 Phase-Shift Continuous-Backend Diagnostics

The following tables and figures document the non-teacher phase-shift continuous-backend evidence. These rows show that some normalized CR3BP phase-shift cases are feasible with research-grade continuous backends, but they do not establish certified trajectory-design capability or quantum evidence.

Table 18: Bounded-projected multiple-shooting diagnostics on the non-teacher catalog halo phase-shift target. Controls are projected to $\|u_i\| \le a_{\max}$ inside the residual and evaluation paths.

Transfer time	amax	Segments	Selected outages	Initialization	Nominal error	Selected recovery worst error	All-outage diagnostic worst error	Fuel	Max ——u——	Bound violation	nfev	Runtime (s)	Meets thresholds
0.5000	0.2000	12	1	linear	0.0479	0.0619	0.0752	0.1000	0.2000	0.0000	24	18.0024	True
0.5000	0.2000	8	8	linear	0.0414	0.3378	0.3378	0.1000	0.2000	0.0000	30	73.0455	False
0.5000	0.2000	12	12	linear	0.0396	0.3904	0.3904	0.1000	0.2000	0.0000	20	141.5146	False

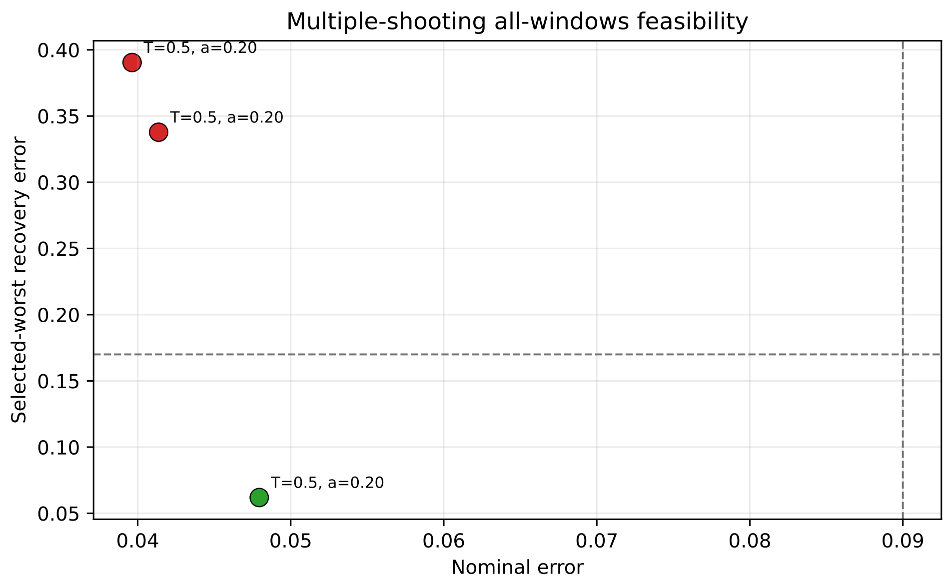

Figure 9: Bounded-projected multiple-shooting diagnostics for the non-teacher phase-shift target.

Table 19: Bounded-projected multiple-shooting phase-suite diagnostics on non-teacher catalog halo phase-shift targets. The selected-branch rows optimize one outage branch; the all-outage-selected row optimizes all one-segment outage branches and remains infeasible.

Case	Phase time	Transfer time	amax	Segments	Selected outages	Nominal error	Selected worst error	All-mask diagnostic worst error	Max ——u——	Bound violation	nfev	Runtime (s)	Meets thresholds
all_outage_selected_diagnostic	0.3000	0.5000	0.2000	12	12	0.0396	0.3581	0.3581	0.2000	0.0000	40	366.6997	False
selected_branch	0.2000	0.5000	0.2000	12	1	0.1939	0.2168	0.2288	0.2000	0.0000	22	12.6848	False
selected_branch	0.3000	0.5000	0.2000	12	1	0.0479	0.0619	0.0752	0.2000	0.0000	24	17.5261	True
selected_branch	0.4000	0.5000	0.2000	12	1	0.0260	0.0083	0.0274	0.2000	0.0000	40	40.3375	True

The all-configured headroom package below is a separate artifact family. It uses the independent-midpoint Hermite–Simpson transcription, warm-starts the original p=0.4, $a_{\max} = 0.2$ row from a p=0.3 cold source, and selects all eight configured one-segment outage masks. The original p=0.4 row reaches nominal error 0.011115 and selected/all worst error 0.077416 under normalized CR3BP, but it stops at `max_nfev=120` with `optimizer_success=False`. The polish row warm-starts from the original p=0.4 row and its persisted branch controls, reaches nominal error 0.011333 and selected/all worst error 0.077921, and converges with `optimizer_success=True` at `nfev=25`. Endpoint and midpoint nominal controls are replayed in the replay/stress table

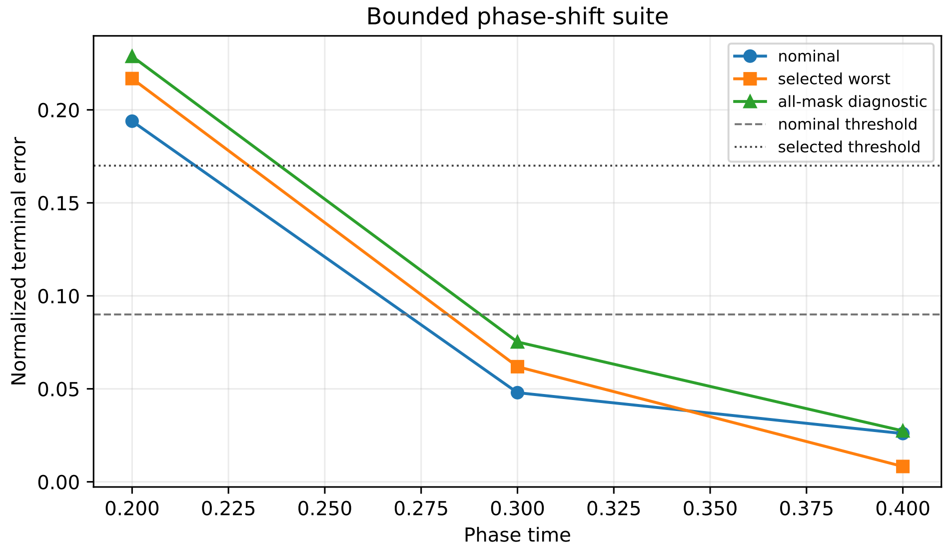

Figure 10: Bounded phase-suite diagnostics on non-teacher catalog halo phase-shift targets.

Table 20: Robust-margin suite on non-teacher catalog halo phase-shift targets. The thrust-margin rows optimize one selected one-segment outage branch and report all one-segment masks diagnostically. The all-single-outage rows optimize every one-segment outage mask for that row; this does not include two-segment outage families.

Group	Phase time	Transfer time	amax	Segments	Max nfev	Selected outages	Outage masks	All selected?	Nominal error	Selected worst error	All-mask worst error	Max ——u——	Bound violation	nfev	Runtime (s)	Meets thresholds
All single outages	0.2000	0.5000	0.3000	8	60	8	8	yes	0.0472	0.2523	0.2523	0.3000	0.0000	60	137.8249	False
All single outages	0.3000	0.5000	0.3000	8	60	8	8	yes	0.0555	0.0358	0.0358	0.3000	0.0000	60	128.1735	True
All single outages	0.4000	0.5000	0.3000	8	60	8	8	yes	0.0531	0.0150	0.0150	0.3000	0.0000	57	119.3214	True
Thrust margin	0.2000	0.5000	0.2000	12	80	1	12	no	0.1939	0.2168	0.2288	0.2000	0.0000	22	11.1493	False
Thrust margin	0.2000	0.5000	0.2500	12	80	1	12	no	0.1136	0.1418	0.1543	0.2500	0.0000	27	14.0958	False
Thrust margin	0.2000	0.5000	0.3000	12	80	1	12	no	0.0538	0.0738	0.0863	0.3000	0.0000	30	16.5662	True

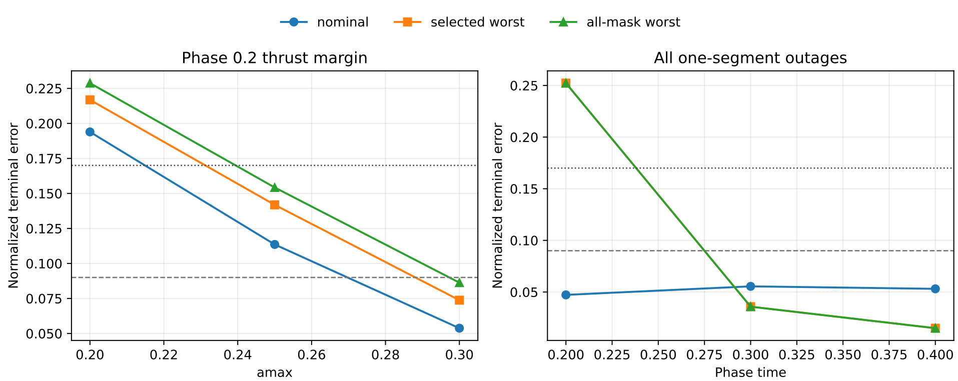

Figure 11: Robust-margin suite summary. The phase-time 0.2 case is thrust-margin sensitive, while higher-thrust smaller cases can meet screening thresholds.

Table 21: Continuation-extension continuous-backend diagnostics on normalized CR3BP catalog halo phase-shift targets. This is a direct multiple-shooting continuation result, not a quantum, QUBO, QAOA, or discrete-sampler result.

Group	Phase time	Warm start	Segments	Outage masks	Max nfev	Nominal error	Selected worst error	All-mask worst error	Optimizer success	MS success	nfev	Runtime (s)	Meets thresholds
all single headroom	0.3000	cold	8	8/8	120	0.0555	0.0351	0.0351	True	True	77	138.7051	True
all single headroom	0.4000	from 0.3	8	8/8	100	0.0531	0.0139	0.0139	True	True	61	120.8934	True
two segment budget test	0.3000	cold	6	11/11	120	0.0592	0.0690	0.0690	True	True	74	106.8293	True
two segment budget test	0.2000	from 0.3	6	11/11	160	0.0651	0.2193	0.2193	False	False	160	253.8218	False
two segment budget test	0.3000	cold	8	15/15	120	0.0612	0.0435	0.0435	True	True	76	243.8341	True

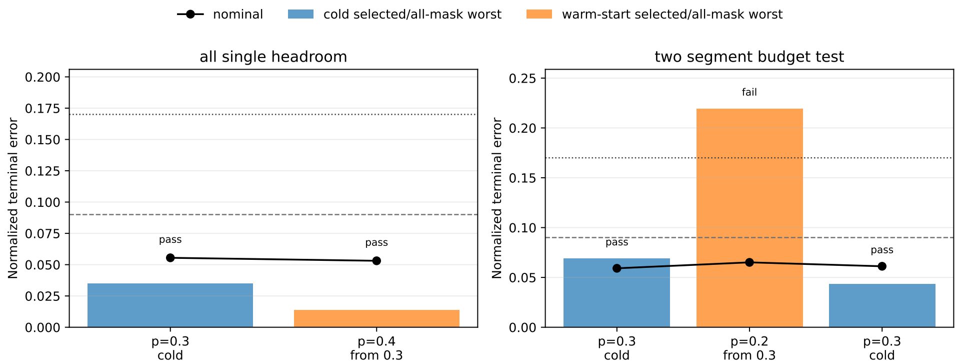

Figure 12: Continuation-extension continuous-backend diagnostics. Warm-start sidecars record nominal-control provenance.

Table 22: Compact bounded projected trapezoidal direct-collocation baseline on the non-teacher catalog halo phase-shift suite. Rows optimize one selected outage branch and report all-mask diagnostics separately.

Case	Phase time	Transfer time	amax	Segments	Selected outages	Nominal error	Selected worst error	All-mask diagnostic worst error	Max ——u——	Bound violation	nfev	Runtime (s)	Meets thresholds
selected_branch	0.2000	0.5000	0.2000	12	1	0.4512	0.4794	0.4805	0.2000	0.0000	30	43.4778	False
selected_branch	0.3000	0.5000	0.2000	12	1	0.3714	0.3774	0.3969	0.2000	0.0000	30	48.8181	False
selected_branch	0.4000	0.5000	0.2000	12	1	0.0357	0.0191	0.0602	0.1483	0.0000	11	18.9927	True

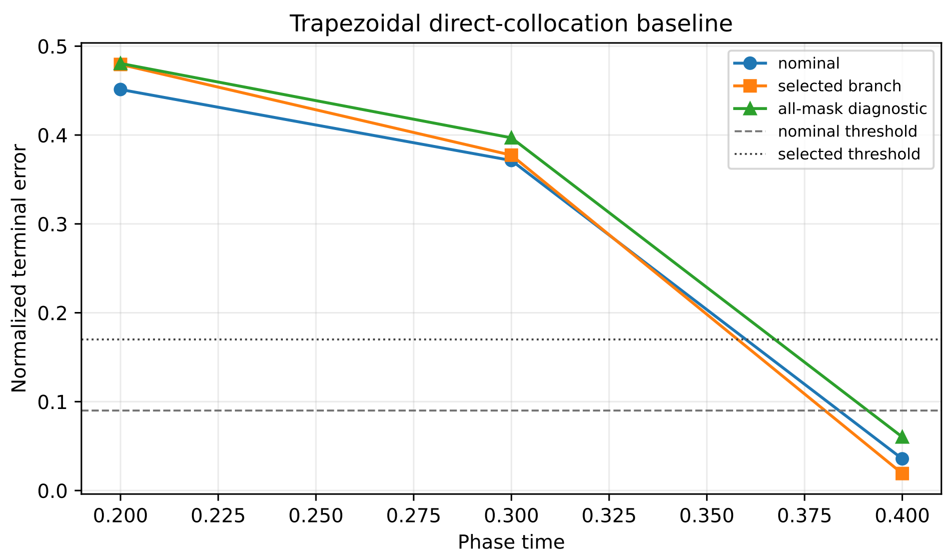

Figure 13: Direct-collocation baseline summary. The compact transcription solves the easiest phase-suite row and fails the harder rows.

Table 23: Constant-control Hermite–Simpson continuation baseline/probe on normalized CR3BP catalog halo phase-shift targets. Warm rows reuse persisted nominal-control sidecars. This is continuous-backend evidence only.

Case ID	Group	Phase time	Warm start	Segments	Outage masks	Max nfev	Method	Nominal error	Selected worst error	All-mask worst error	Optimizer success	DC success	nfev	Runtime (s)	Meets thresholds
hs_phase_p03_cold	Phase continuation (single outage)	0.3000	cold	8	3/8	50	hermite_simpson	0.0376	0.0287	0.0901	False	True	50	145.1586	True
hs_phase_p02_warm_from_p03	Phase continuation (single outage)	0.2000	from 0.3	8	3/8	80	hermite_simpson	0.1069	0.1208	0.1791	False	False	80	249.4395	False
hs_phase_p04_warm_from_p03	Phase continuation (single outage)	0.4000	from 0.3	8	3/8	80	hermite_simpson	0.0179	0.0128	0.0628	True	True	7	25.1659	True
hs_hard_p04_amax02_warm_from_p03	Hard catalog stress probe	0.4000	from 0.3	8	3/8	60	hermite_simpson	0.0183	0.0176	0.0439	True	True	24	74.8323	True

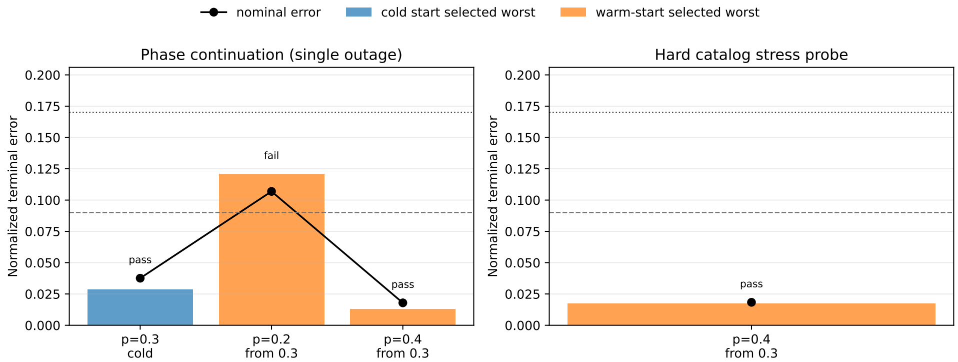

Figure 14: Constant-control Hermite–Simpson continuation baseline/probe summary.

Table 24: Independent-midpoint-control Hermite–Simpson continuation evidence. Warm rows reuse persisted endpoint and midpoint nominal-control sidecars when available. This is continuous-backend evidence only; the catalog-DRO row is a bounded selected-outage negative diagnostic.

Case ID	Group	Phase time	Warm start	Segments	Outage masks	Max nfev	Method	Nominal error	Selected worst error	All-mask worst error	Optimizer success	DC success	nfev	Runtime (s)	Meets thresholds
ihs_phase_p03_cold	Phase continuation (single outage)	0.3000	cold	8	3/8	60	hermite_simpson_midpoint	0.0358	0.0300	0.0849	False	True	60	221.3432	True
ihs_phase_p02_warm_from_p03	Phase continuation (single outage)	0.2000	from 0.3	8	3/8	90	hermite_simpson_midpoint	0.0976	0.0942	0.1694	False	False	90	419.5509	False
ihs_phase_p04_warm_from_p03	Phase continuation (single outage)	0.4000	from 0.3	8	3/8	90	hermite_simpson_midpoint	0.0194	0.0152	0.0605	True	True	7	34.3690	True
ihs_phase_p04_amax02_warm_from_p03	tighter thrust phase shift	0.4000	from 0.3	8	3/8	80	hermite_simpson_midpoint	0.0197	0.0188	0.0532	True	True	17	73.7907	True
ihs_catalog_dro_tf4_amax1_selected1	hard catalog selected outage	0.0000	cold	8	1/15	50	hermite_simpson_midpoint	4.2644	11.0843	11.0843	False	False	50	50.2186	False

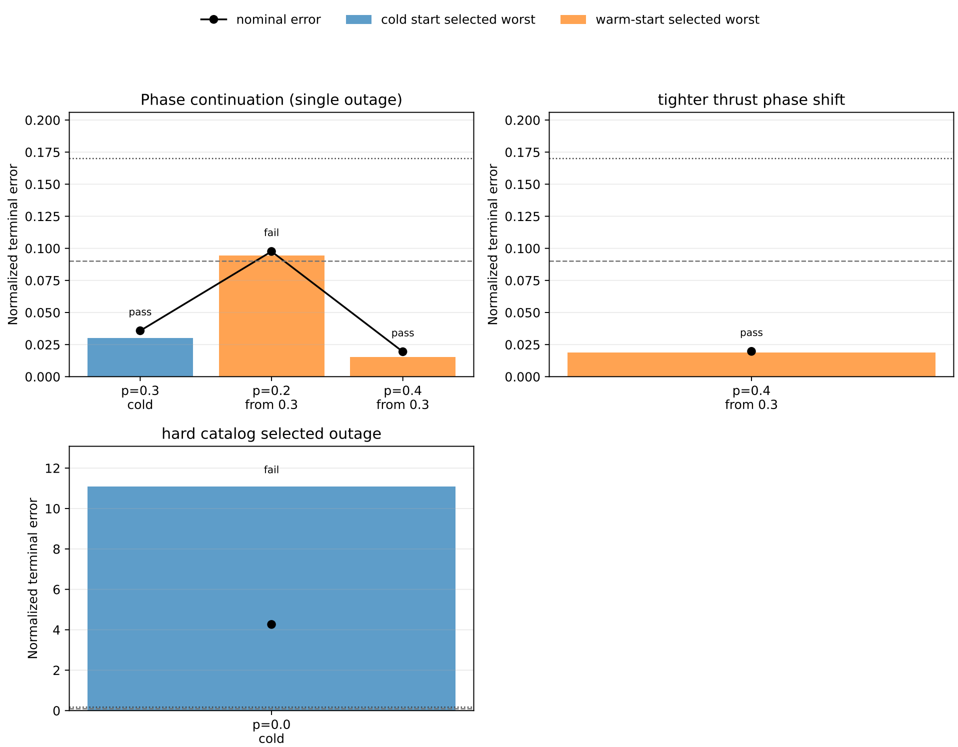

Figure 15: Independent-midpoint-control Hermite–Simpson continuation summary. The package strengthens the continuous-backend comparison but keeps the phase-time 0.2 and catalog-DRO diagnostics unresolved.

at the source grid, refined grids, and direct acceleration scales. The separate branch-control replay table validates the p=0.4 manifests and sidecar hashes, then repropagates 16 persisted branch-control schedules with zero deltas at tolerance 10^{-10}. These are normalized-CR3BP replay and continuation artifacts only, not broad production-solver validation/parity or high-fidelity validation.

Table 25: Independent-midpoint-control Hermite–Simpson all-configured one-segment headroom package. This is normalized-CR3BP continuous-backend evidence only, not broad production-solver validation/parity or high-fidelity validation.

Case ID	Group	Phase time	Warm start	Segments	Outage masks	Max nfev	Method	Nominal error	Selected worst error	All-mask worst error	Optimizer success	DC success	nfev	Runtime (s)	Meets thresholds
ihs_source_p03_cold	phase all configured single	0.3000	cold	8	3/8	60	hermite_simpson_midpoint	0.0358	0.0300	0.0849	False	True	60	251.0253	True
ihs_all_single_p04_amax02_warm_from_p03	phase all configured single	0.4000	from 0.3	8	8/8	120	hermite_simpson_midpoint	0.0111	0.0774	0.0774	False	True	120	2685.2621	True
ihs_all_single_p04_amax02_polish_from_p04	phase all configured single	0.4000	from 0.4	8	8/8	240	hermite_simpson_midpoint	0.0113	0.0779	0.0779	True	True	25	497.3724	True

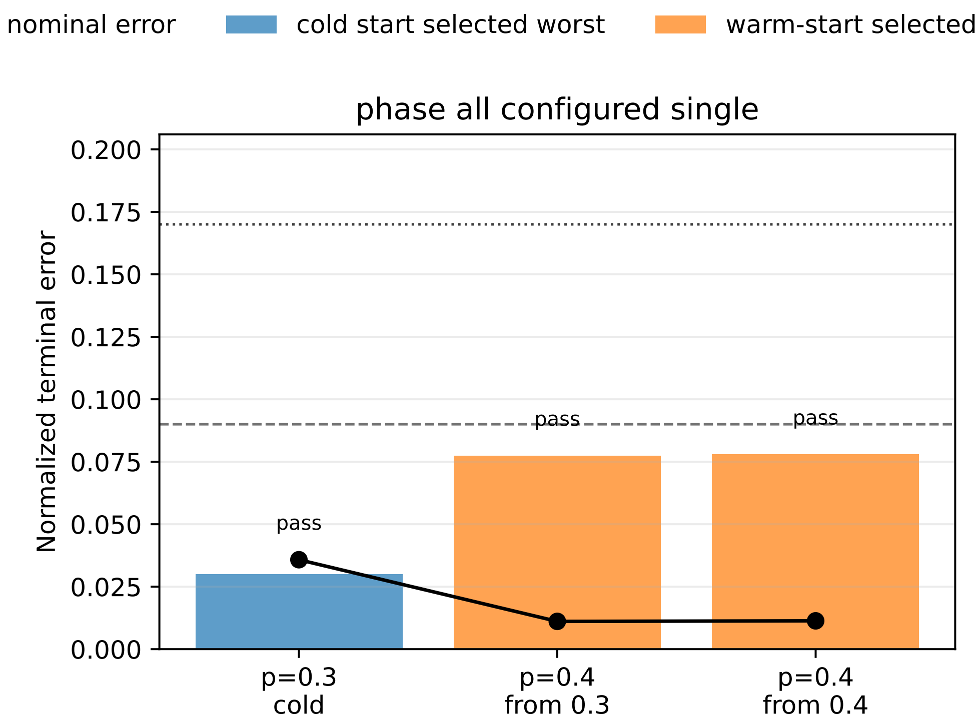

Figure 16: Independent-midpoint-control Hermite–Simpson all-configured headroom summary for the p=0.4, $a_{\max} = 0.2$ phase-shift row.

The simple bicircular phase-sweep stress package reuses the replay-ready p=0.4 independent-HS endpoint and midpoint sidecars. It validates the branch manifests and sidecar SHA-256 hashes, reproduces the CR3BP recorded errors with maximum delta 0.0, and then repropagates the same controls with a circular solar third-body tidal term at Sun phases $0°, 45°, \ldots, 315°$. For the converged polish row, all 8 nominal phase rows and all 64 branch-phase rows pass the configured 0.09/0.17 thresholds. The maximum simple-bicircular nominal error is 0.022138676654057693, and the maximum branch-phase error is 0.08557051343145317. This is a deterministic stress-probe result only; it is not SPICE ephemeris validation, high-fidelity flight validation, broad production-solver validation/parity, fuel optimality, or quantum evidence.

The cached-Horizons-derived independent-HS solar-tidal replay package reads the committed cache in `data/cache/horizons/independent_hs_phase_shift_2026jan01_vectors.json`. The

Table 26: Independent-HS simple bicircular phase-sweep stress replay from persisted endpoint-plus-midpoint sidecars. The converged polish row passes all tested phases and branch-phase checks under the configured thresholds, but the model is a simple circular solar-tidal stress probe only.

Case	Phases	Max nominal	Max branch	Nominal pass	Nominal rows	Branch pass	Branch rows	All pass
ihs_all_single_p04_amax02_polish_from_p04	8	0.0221387	0.0855705	8	8	64	64	True
ihs_all_single_p04_amax02_warm_from_p03	8	0.0216141	0.085064	8	8	64	64	True

cache stores JPL Horizons Moon and Sun geocentric vectors at the nine p=0.4 independent-HS segment nodes from the representative 2026-Jan-01 TDB epoch, then converts the Sun geometry to the Earth–Moon rotating barycentric frame and linearly interpolates the Sun vector over canonical time. The replay reuses the persisted endpoint-plus-midpoint controls and the same fixed CR3BP target. For the converged polish row, the cached-Horizons-derived solar-tidal nominal error is 0.018363195236986728, the branch worst error is 0.07422350563850917, the branch pass count is 8/8, the CR3BP replay delta is 0.0, and the Sun-distance range is 382.6857920288508–382.693044178952 LU. This is stronger than the simple circular phase sweep because it uses cached JPL Horizons geometry at a representative epoch, but it is still a simplified stress probe, not SPICE propagation, high-fidelity or flight validation, flight-ready evidence, or broad production-solver validation/parity.

Table 27: Independent-HS cached-Horizons-derived solar-tidal replay from persisted endpoint-plus-midpoint sidecars at a representative 2026-Jan-01 epoch. The converged polish row passes the configured thresholds, but the model remains a simplified stress probe rather than SPICE/high-fidelity validation or broad production-solver validation/parity.

Case	Horizons nominal	Horizons branch worst	Branch pass	Branch rows	CR3BP replay delta	Sun LU min	Sun LU max	Cache SHA-256
ihs_all_single_p04_amax02_polish_from_p04	0.0183632	0.0742235	8	8	0	382.686	382.693	13fe699371ad
ihs_all_single_p04_amax02_warm_from_p03	0.018381	0.0737478	8	8	0	382.686	382.693	13fe699371ad

The cached-Horizons Earth/Moon/Sun point-mass retuning package uses the same committed cache, representative 2026-Jan-01 epoch, fixed final time, and p=0.4 independent-HS endpoint-plus-midpoint control semantics. It converts the CR3BP rotating barycentric states to geocentric inertial coordinates using the cached Moon vector, propagates Earth central gravity plus indirect Moon and Sun point-mass terms, and transforms controls into the inertial frame at RK stages. Direct replay of the persisted polish controls fails the configured point-mass thresholds: nominal replay error is 0.3812580376880591, persisted branch worst error is 0.3797450961017463, and persisted branch pass count is 0/8. The script then reruns independent least-squares retuning from the persisted nominal and branch endpoint-plus-midpoint controls, with projected controls at $a_{\max} = 0.2$ and modest smooth/control regularization. Retuning gives nominal error 0.02143944130524006, branch worst error 0.02473065115224942, branch pass count 8/8, optimizer success count 9/9, and total function evaluations 71. The cache SHA-256 is `13fe699371ad67bf1616d38b7afd316bbff72811bbc0f8337cff51d6333897b2`. This is a cached-Horizons point-mass stress/retuning model only; it is not SPICE propagation, full high-fidelity or flight validation, broad production-solver validation/parity, fuel optimality, DOI evidence, or quantum evidence.

The multi-epoch cached-Horizons Earth/Moon/Sun point-mass retuning wrapper addresses the representative-epoch concern by repeating the same persisted-control replay and independent retuning protocol at four fixed 2026 epochs: 2026-Jan-01, 2026-Apr-01, 2026-Jul-01, and 2026-Oct-01. The aggregate artifacts are `data/results/independent_hs_horizons_multi_epoch_point_mass_retuning/independent_hs_horizons_multi_epoch_`

Table 28: Independent-HS cached-Horizons Earth/Moon/Sun point-mass retuning at a representative 2026-Jan-01 epoch. Persisted controls fail direct replay under the point-mass model, but independent retuning restores nominal and all 8 branch checks below the configured thresholds. This is retuning evidence, not SPICE/high-fidelity validation or broad production-solver validation/parity.

Case	Persisted nominal replay	Retuned nominal	Persisted branch worst	Retuned branch worst	Branch pass	Cache SHA-256
ihs_all_single_p04_amax02_polish_from_p04	0.381258	0.0214394	0.379745	0.0247307	8/8	13fe699371ad

`point_mass_retuning.csv`, `data/results/independent_hs_horizons_multi_epoch_point_mass_retuning/independent_hs_horizons_multi_epoch_point_mass_retuning_metadata.json`, per-epoch outputs under `data/results/independent_hs_horizons_multi_epoch_point_mass_retuning/epochs/`, and the table below. The default run is offline from committed cache files `data/cache/horizons/independent_hs_phase_shift_2026jan01_vectors.json`, `data/cache/horizons/independent_hs_phase_shift_2026apr01_vectors.json`, `data/cache/horizons/independent_hs_phase_shift_2026jul01_vectors.json`, and `data/cache/horizons/independent_hs_phase_shift_2026oct01_vectors.json`. Across 36 rows (4 nominal and 32 branch rows), nominal direct replay fails in all four epochs; direct branch pass count is 18/32 overall, including 8/8 in July. Worst direct nominal and branch errors are 0.3812580376880591 and 0.3797450961017463. Retuning restores feasibility for every row: worst retuned nominal and branch errors are 0.02143944130524006 and 0.02473065115224942, branch pass count is 32/32, SciPy success count is 36/36, and total function evaluations are 197. This is a stronger representative-epoch point-mass stress/retuning package, but it is still not SPICE propagation, full high-fidelity or flight validation, broad production-solver validation/parity, fuel optimality, DOI evidence, or quantum evidence.

Table 29: Independent-HS multi-epoch cached-Horizons Earth/Moon/Sun point-mass retuning. Each epoch first replays persisted controls under the point-mass model and then independently retunes nominal and branch controls. Nominal direct replay fails in every epoch; all retuned rows pass the configured thresholds.

Epoch	Start JD TDB	Direct nominal	Direct branch worst	Direct branch pass	Retuned nominal	Retuned branch worst	Retuned branch pass	SciPy success	nfev	Cache SHA-256
2026-Jan-01	2.46104e+06	0.381258	0.379745	0/8	0.0214394	0.0247307	8/8	9/9	71	13fe699371ad
2026-Apr-01	2.46113e+06	0.107365	0.179059	7/8	0.0100707	0.0191138	8/8	9/9	58	83f859c622aa
2026-Jul-01	2.46122e+06	0.0990305	0.141091	8/8	0.0128722	0.0194392	8/8	9/9	39	5a5e2d844f3b
2026-Oct-01	2.46131e+06	0.189591	0.187878	3/8	0.0144302	0.0180131	8/8	9/9	29	debc9260f744

The SPICE-derived ephemeris replay package uses the four same canonical epoch grids as the multi-epoch Horizons wrapper, but the committed caches in `data/cache/spice/` are derived from NAIF SPICE kernels `naif0012.tls`, `de442s.bsp`, and `gm_de440.tpc`. The cache metadata records kernel URLs, kernel SHA-256 values, SpiceyPy/CSPICE versions, `J2000` frame, `NONE` aberration correction, Earth observer, Moon/Sun targets, TDB calendar labels, and the source limitation that these are geometric vector samples for a point-mass replay. The replay does not rerun optimization: it reads the existing multi-epoch Horizons-retuned nominal and one-segment branch control sidecars, reconstructs the same endpoint-plus-midpoint control semantics, and repropagates those 36 controls under the SPICE-derived Earth/Moon/Sun point-mass profile. All 4 nominal rows and all 32 branch rows pass their configured thresholds. Worst SPICE replay nominal and branch errors are 0.021439441253166033 and 0.024730650824609506, and the maximum absolute delta from the corresponding Horizons-retuned replay is $3.2763991519857427 \times 10^{-10}$. This is a scoped ephemeris-source replay under the same point-mass force model, not full high-fidelity or flight validation, broad production-solver validation/parity, fuel optimality, DOI evidence, or quantum evidence.

Table 30: Independent-HS SPICE-derived ephemeris replay of the multi-epoch Horizons-retuned controls. The package replays already-retuned controls under compact SPICE-derived Moon/Sun vector caches without retuning or optimization rerun; it is point-mass ephemeris-source replay, not full high-fidelity validation.

Epoch	SPICE nominal	SPICE branch worst	SPICE branch pass	Horizons-retuned nominal	Horizons-retuned branch	Max delta	Cache SHA-256
2026-Jan-01	0.0214394	0.0247307	8/8	0.0214394	0.0247307	3.2764e-10	fd785770a77c
2026-Apr-01	0.0100707	0.0191138	8/8	0.0100707	0.0191138	1.70991e-10	2792bd47aeb8
2026-Jul-01	0.0128722	0.0194392	8/8	0.0128722	0.0194392	1.04669e-10	1e5631b4574d
2026-Oct-01	0.0144302	0.0180131	8/8	0.0144302	0.0180131	1.00527e-10	47e79d632c98

The CasADi/IPOPT bridge package addresses the mature-backend portability/bridge concern in a deliberately narrow normalized-CR3BP setting. It reads the accepted `ihs_all_single_p04_amax02_polish_fro` nominal sidecar and eight branch sidecars, verifies the source manifest and sidecar SHA-256 hashes, and exports nominal plus post-recovery active branch endpoint-plus-midpoint controls to a CasADi direct-shooting NLP solved with IPOPT. Branch outage-masked segments remain fixed inactive/zero, and each branch pre-recovery prefix is fixed to the refined nominal controls with the outage mask applied, so the target, scales, configured thresholds, and branch-recovery semantics match the independent-HS source artifacts. Source CR3BP replay gives nominal/branch-worst errors 0.011333095366088189/0.07792080291839382; IPOPT refinement gives 0.009138565365046585/0.015534969964216154, with IPOPT success 9/9, bridge pass 9/9, total IPOPT iterations 66, maximum prefix delta 0.0, and maximum control-bound violation 0.0. This is a scoped CasADi/IPOPT mature NLP backend bridge check only; it is not production mission design, high-fidelity or flight validation, global or fuel optimality, DOI evidence, or quantum evidence.

Table 31: Independent-HS CasADi/IPOPT bridge check for the accepted normalized-CR3BP polish case. The bridge fixes branch pre-recovery prefixes to refined nominal masked controls and refines only post-recovery active endpoint-plus-midpoint controls under the same target/scales/branch semantics; it is scoped to a mature NLP backend bridge check, not production mission design or high-fidelity validation.

Type	Mask	Recovery start	Source replay	IPOPT refined	Threshold	Bridge pass	IPOPT success	Iterations
nominal			0.0113331	0.00913857	0.09	True	True	9
branch	0	1	0.0779208	0.0146545	0.17	True	True	10
branch	1	2	0.0557935	0.0129775	0.17	True	True	9
branch	2	3	0.0379084	0.0113004	0.17	True	True	7
branch	3	4	0.0250732	0.00990266	0.17	True	True	7
branch	4	5	0.0175256	0.00906888	0.17	True	True	7
branch	7	8	0.00971579	0.015535	0.17	True	True	0
branch	6	7	0.0105362	0.00909914	0.17	True	True	9
branch	5	6	0.0133582	0.00884673	0.17	True	True	8

6 Phase-Shift Schedule Diagnostics and Small-Seed Evidence

The unrepaired $N = 8$ phase-shift benchmark, deterministic dense-repair ablation, and legacy 10-seed duty-cycle-prior package explain the schedule-encoding failure mode behind the main 30-seed result. The unrepaired samplers can choose sparse late-thrust schedules that do not refine, while dense repair collapses methods to the same all-windows envelope. The 10-seed cardinality-prior package is retained as historical small-seed evidence; the main article uses the 30-seed package for claims.

Table 32: Catalog halo phase-shift benchmark. The target is generated by ballistic CR3BP phase propagation of the JPL source state, not by a low-thrust teacher.

Benchmark	Method	Comparison group	Refined nominal error	Selected recovery worst error	All-mask diagnostic worst error	Active fraction	Solver true evals	Shared QUBO train evals	Total true evals	Refine success
catalog_halo_phase_shift	random	method_comparison	0.2663	0.3042	0.3042	0.3750	52.0000	0.0000	52.0000	0.0000
catalog_halo_phase_shift	cross_entropy	method_comparison	0.2663	0.3042	0.3042	0.3750	56.0000	0.0000	56.0000	0.0000
catalog_halo_phase_shift	genetic	method_comparison	0.2663	0.3042	0.3042	0.3750	52.0000	0.0000	52.0000	0.0000
catalog_halo_phase_shift	true_sa	method_comparison	0.2663	0.3042	0.3042	0.3750	52.0000	0.0000	52.0000	0.0000
catalog_halo_phase_shift	surrogate_qubo_sa	method_comparison	0.2663	0.3042	0.3042	0.3750	20.0000	32.0000	52.0000	0.0000
catalog_halo_phase_shift	qaoa_statevector	method_comparison	0.2242	0.2625	0.2650	0.5000	20.0000	32.0000	52.0000	0.0000
catalog_halo_phase_shift	all_windows_continuous	method_comparison	0.0418	0.0750	0.0887	1.0000	30.0000	0.0000	30.0000	1.0000

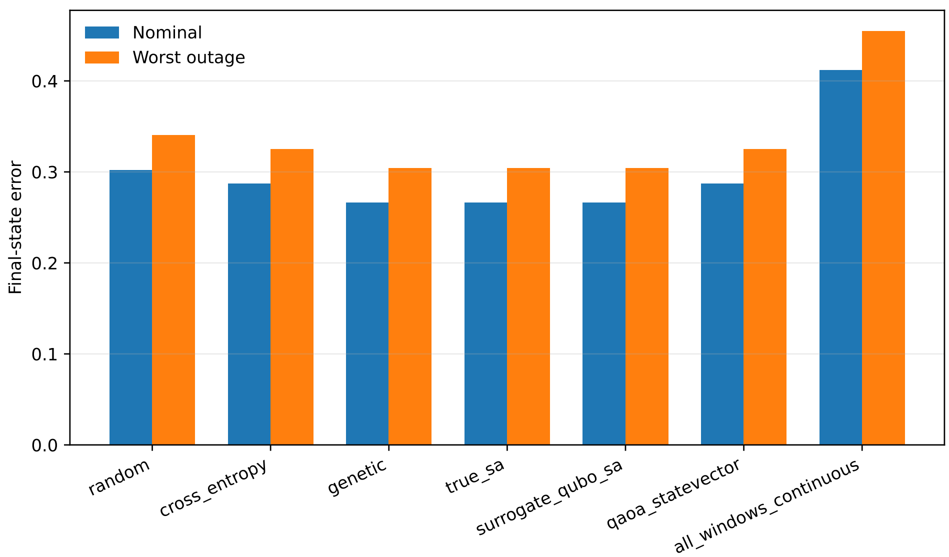

Figure 17: Unrepaired phase-shift method comparison. The all-windows continuous baseline succeeds where sparse binary schedules do not.

Table 33: Catalog halo phase-shift benchmark with deterministic dense-schedule repair. The table exposes the solver schedule, refinement candidate schedule, repaired refined schedule, and active fractions to show that repaired methods collapse to the same all-windows refinement envelope.

Benchmark	Method	Schedule repair	Refined schedule source	Comparison group	Solver best schedule	Refine candidate schedule	Refined schedule	Refined nominal error	Selected recovery worst error	All-mask diagnostic worst error	Solver best active fraction	Refine candidate active fraction	Refined active fraction	Solver true evals	Shared QUBO train evals	Total true evals	Refine success
catalog_halo_phase_shift	random	prefix_to_last_active	heuristic dense-availability envelope: set every window through the last active solver window to available, preserving later zeros	method_comparison	mixed	mixed	11111111	0.0418	0.0750	0.0887	0.3750	0.3750	1.0000	52.0000	0.0000	52.0000	1.0000
catalog_halo_phase_shift	cross_entropy	prefix_to_last_active	heuristic dense-availability envelope: set every window through the last active solver window to available, preserving later zeros	method_comparison	mixed	mixed	11111111	0.0418	0.0750	0.0887	0.3750	0.3750	1.0000	56.0000	0.0000	56.0000	1.0000
catalog_halo_phase_shift	genetic	prefix_to_last_active	heuristic dense-availability envelope: set every window through the last active solver window to available, preserving later zeros	method_comparison	00000111	00000111	11111111	0.0418	0.0750	0.0887	0.3750	0.3750	1.0000	52.0000	0.0000	52.0000	1.0000
catalog_halo_phase_shift	true_sa	prefix_to_last_active	heuristic dense-availability envelope: set every window through the last active solver window to available, preserving later zeros	method_comparison	mixed	mixed	11111111	0.0418	0.0750	0.0887	0.3750	0.3750	1.0000	52.0000	0.0000	52.0000	1.0000
catalog_halo_phase_shift	surrogate_qubo_sa	prefix_to_last_active	heuristic dense-availability envelope: set every window through the last active solver window to available, preserving later zeros	method_comparison	00000111	00000111	11111111	0.0418	0.0750	0.0887	0.3750	0.3750	1.0000	20.0000	32.0000	52.0000	1.0000
catalog_halo_phase_shift	qaoa_statevector	prefix_to_last_active	heuristic dense-availability envelope: set every window through the last active solver window to available, preserving later zeros	method_comparison	mixed	mixed	11111111	0.0418	0.0750	0.0887	0.5000	0.5000	1.0000	20.0000	32.0000	52.0000	1.0000
catalog_halo_phase_shift	all_windows_continuous	identity	original solver candidate schedule	method_comparison	11111111	11111111	11111111	0.0418	0.0750	0.0887	1.0000	1.0000	1.0000	30.0000	0.0000	30.0000	1.0000

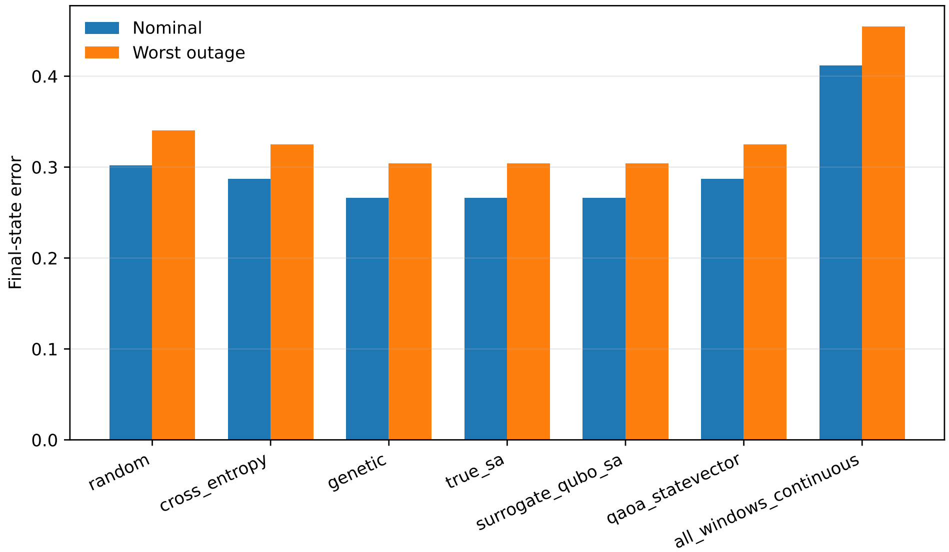

Figure 18: Dense-repair phase-shift comparison. The repair is an encoding diagnostic, not a competitive method-ranking result.

Table 34: Legacy 10-seed $N = 12$ duty-cycle-prior phase-shift benchmark. QUBO/QAOA totals include 96 shared QUBO-training evaluations and 24 post-QUBO true candidate evaluations.

Benchmark	Method	Schedule repair	Refined schedule source	Comparison group	Solver best schedule	Refine candidate schedule	Refined schedule	Refined nominal error	Selected recovery worst error	All-mask diagnostic worst error	Solver best active fraction	Active target penalty	Refine candidate active fraction	Refined active fraction	Solver true evals	Shared QUBO train evals	Total true evals	Refine success
catalog_halo_phase_shift	random	identity	original solver candidate schedule	method_comparison	mixed	mixed	mixed	0.0954	0.1214	0.1313	0.8333	0.0833	0.8333	0.8333	120.0000	0.0000	120.0000	0.2000
catalog_halo_phase_shift	cross_entropy	identity	original solver candidate schedule	method_comparison	mixed	mixed	mixed	0.0690	0.0941	0.1018	0.9167	0.0000	0.9167	0.9167	120.0000	0.0000	120.0000	1.0000
catalog_halo_phase_shift	genetic	identity	original solver candidate schedule	method_comparison	mixed	mixed	mixed	0.0675	0.0926	0.1020	0.9167	0.0000	0.9167	0.9167	120.0000	0.0000	120.0000	1.0000
catalog_halo_phase_shift	true_sa	identity	original solver candidate schedule	method_comparison	mixed	mixed	mixed	0.0690	0.0941	0.1018	0.9167	0.0000	0.9167	0.9167	120.0000	0.0000	120.0000	1.0000
catalog_halo_phase_shift	surrogate_qubo_sa	identity	original solver candidate schedule	method_comparison	mixed	mixed	mixed	0.0690	0.0941	0.1018	0.9167	0.0000	0.9167	0.9167	24.0000	96.0000	120.0000	1.0000
catalog_halo_phase_shift	qaoa_statevector	identity	original solver candidate schedule	method_comparison	mixed	mixed	mixed	0.0942	0.1201	0.1292	0.8333	0.0833	0.8333	0.8333	24.0000	96.0000	120.0000	0.4000
catalog_halo_phase_shift	all_windows_continuous	identity	original solver candidate schedule	method_comparison	111111111111	111111111111	111111111111	0.0386	0.0601	0.0690	1.0000	0.0833	1.0000	1.0000	19.0000	0.0000	19.0000	1.0000

Table 35: Legacy 10-seed duty-cycle-prior recovery fuel and refinement effort. Fuel is reported in normalized acceleration-time units.

Method	Schedule repair	Refined schedule	Refined schedule source	Nominal fuel	Mean recovery fuel	Max recovery fuel	Refinement nfev	Recovery success
random	identity	mixed	original solver candidate schedule	0.0833	0.0750	0.0750	26.0000	0.2000
cross_entropy	identity	mixed	original solver candidate schedule	0.0917	0.0833	0.0833	20.0000	1.0000
genetic	identity	mixed	original solver candidate schedule	0.0917	0.0833	0.0833	35.0000	1.0000
true_sa	identity	mixed	original solver candidate schedule	0.0917	0.0833	0.0833	20.0000	1.0000
surrogate_qubo_sa	identity	mixed	original solver candidate schedule	0.0917	0.0833	0.0833	20.0000	1.0000
qaoa_statevector	identity	mixed	original solver candidate schedule	0.0833	0.0750	0.0750	27.0000	0.4000
all_windows_continuous	identity	111111111111	original solver candidate schedule	0.1000	0.0917	0.0917	19.0000	1.0000

Table 36: Legacy 10-seed duty-cycle-prior QUBO surrogate diagnostics.

seed	ridge	train_rmse	val_rmse	train_r2	val_r2	val_spearman	val_pairwise_order_accuracy	val_top_k	val_top_k_recall	n_train	n_val	shared_qubo_training_evaluations	qubo_training_true_evaluations	equal_total_budget
11	0.0001	0.0001	0.1056	1.0000	0.9961	0.9896	0.9674	5	1.0000	72	24	96	96	True
23	0.0001	0.0001	0.5663	1.0000	0.9220	0.9000	0.8804	5	0.6000	72	24	96	96	True
37	0.0001	0.0001	0.1579	1.0000	0.9923	0.9878	0.9638	5	1.0000	72	24	96	96	True
41	0.0001	0.0002	0.5265	1.0000	0.9405	0.9470	0.9130	5	0.8000	72	24	96	96	True
53	0.0001	0.0001	0.2576	1.0000	0.9601	0.9487	0.9275	5	1.0000	72	24	96	96	True
67	0.0001	0.0001	0.4435	1.0000	0.9677	0.9652	0.9348	5	0.6000	72	24	96	96	True
79	0.0001	0.0001	0.2407	1.0000	0.9848	0.9756	0.9491	5	0.8000	72	24	96	96	True
83	0.0001	0.0001	0.2601	1.0000	0.9744	0.9557	0.9275	5	0.8000	72	24	96	96	True
97	0.0001	0.0001	0.1768	1.0000	0.9916	0.9791	0.9529	5	1.0000	72	24	96	96	True
101	0.0001	0.0001	0.2653	1.0000	0.9591	0.9748	0.9384	5	1.0000	72	24	96	96	True

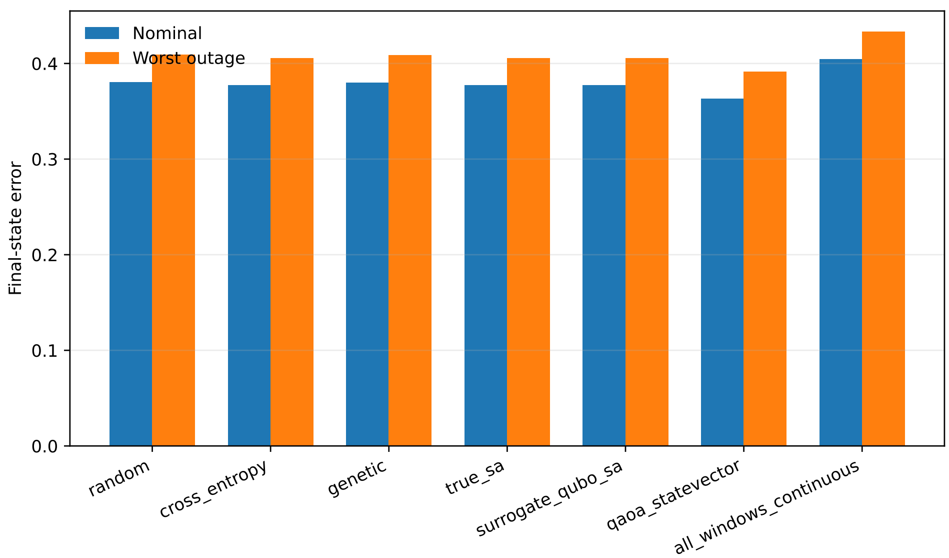

Figure 19: Legacy 10-seed duty-cycle-prior method comparison. The main article uses the 30-seed package for statistical claims.

7 Cardinality and Fuel Ablations

The duty-cycle-prior result is positive but narrow. The high-duty ablations show that one-coast schedules belong to a broad feasible region, and the fuel sweep shows that stronger all-windows continuous fuel regularization can reduce fuel while retaining feasibility. These checks prevent overclaiming one-coast schedules as Pareto-superior.

Table 37: Exhaustive high-duty cardinality ablation for the $N = 12$ phase-shift benchmark.

k_active	schedules_refined	success_count	all_mask_success_count	best_nominal_error	median_nominal_error	best_selected_worst_error	median_selected_worst_error	best_all_mask_worst_error	median_all_mask_worst_error	best_nominal_fuel	median_nominal_fuel	best_schedule_by_selected_worst	min_fuel_feasible_schedule
10	66	41	41	0.0798	0.0874	0.1056	0.1132	0.1146	0.1230	0.0833	0.0833	111111111100	111111011110
11	12	12	12	0.0588	0.0617	0.0833	0.0864	0.0923	0.0956	0.0917	0.0917	111111111110	111111111011
12	1	1	1	0.0386	0.0386	0.0601	0.0601	0.0690	0.0690	0.1000	0.1000	111111111111	111111111111

Table 38: One-coast method selections compared with the refined one-coast frontier.

method	runs	one_coast_runs	one_coast_success_runs	one_coast_frontier_runs	near_best_selected_runs	median_method_selected_worst_error	median_method_nominal_fuel	median_exhaustive_selected_worst_error	median_exhaustive_nominal_fuel	median_selected_gap_to_best_one_coast	dominant_one_coast_schedules
surrogate_qubo_sa	10	10	10	0	0	0.0941	0.0917	0.0941	0.0917	0.0108	011111111111:7; 101111111111:3
cross_entropy	10	10	10	0	1	0.0941	0.0917	0.0941	0.0917	0.0108	011111111111:6; 110111111111:2; 111110111111:1; 111011111111:1
genetic	10	10	10	0	1	0.0926	0.0917	0.0926	0.0917	0.0093	101111111111:5; 110111111111:3; 011111111111:1; 111101111111:1
true_sa	10	10	10	0	0	0.0941	0.0917	0.0941	0.0917	0.0108	011111111111:6; 101111111111:4
qaoa_statevector	10	4	4	1	2	0.0887	0.0917	0.0887	0.0917	0.0054	111011111111:2; 111101111111:1; 111111111110:1
random	10	1	1	0	0	0.0926	0.0917	0.0926	0.0917	0.0093	101111111111:1

Table 39: One-coast refined schedule diagnostics for the duty-cycle-prior ablation.

schedule	coast_windows_zero_based	refinement_success	one_coast_pareto_frontier	refined_nominal_error	refined_selected_worst_error	refined_all_mask_worst_error	refined_nominal_fuel	refinement_nfev
011111111111	0	True	False	0.0690	0.0941	0.1018	0.0917	20
101111111111	1	True	False	0.0675	0.0926	0.1022	0.0917	35
110111111111	2	True	False	0.0670	0.0922	0.1015	0.0917	35
111011111111	3	True	False	0.0643	0.0892	0.0986	0.0917	35
111101111111	4	True	False	0.0633	0.0881	0.0975	0.0917	35
111110111111	5	True	False	0.0628	0.0876	0.0967	0.0917	35
111111011111	6	True	False	0.0601	0.0837	0.0941	0.0917	16
111111101111	7	True	False	0.0606	0.0852	0.0944	0.0917	35
111111110111	8	True	False	0.0601	0.0847	0.0938	0.0917	35
111111111011	9	True	True	0.0603	0.0848	0.0939	0.0917	35
111111111101	10	True	True	0.0596	0.0841	0.0931	0.0917	35
111111111110	11	True	True	0.0588	0.0833	0.0923	0.0917	35

Table 40: All-windows fuel-regularization sweep for the $N = 12$ phase-shift benchmark.

fuel_residual_weight	refinement_success	all_mask_success	refined_nominal_error	refined_selected_worst_error	refined_all_mask_worst_error	refined_nominal_fuel	refinement_nfev
0.0180	True	True	0.0386	0.0601	0.0690	0.1000	19
0.0300	True	True	0.0468	0.0711	0.0791	0.0962	18
0.0500	True	True	0.0773	0.1100	0.1122	0.0846	15
0.0800	False	False	0.1204	0.1491	0.1564	0.0700	10
0.1200	False	False	0.1891	0.2128	0.2183	0.0504	7
0.2000	False	False	0.2755	0.2892	0.2915	0.0268	7

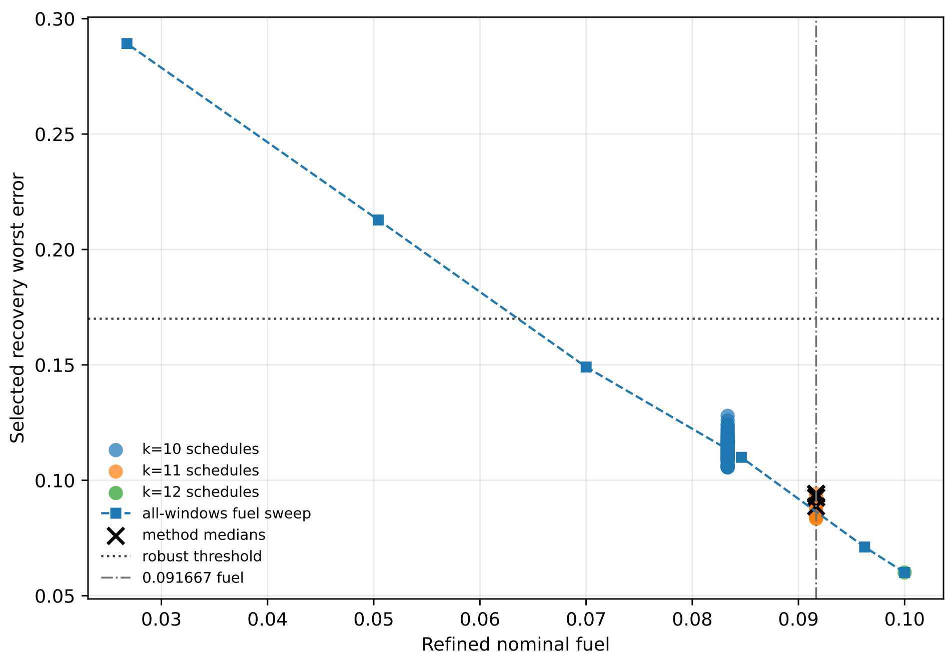

Figure 20: Fuel-error Pareto view for the duty-cycle-prior ablation. Lower fuel is achievable with all-windows fuel regularization while retaining feasibility, so the one-coast result should be read as a schedule-structure feasibility result rather than a fuel-optimality result.

8 Focused QAOA/QUBO Raw Summary and Legacy QAOA Artifacts

The main article reports the 30-seed QAOA/QUBO statistical diagnostics. The raw 30-seed method summary is retained here. Legacy 10-seed QAOA-depth artifacts remain in the repository for provenance but are not the current statistical evidence.

Table 41: QAOA depth and angle-optimization ablation on the $N = 12$ duty-cycle-prior phase-shift benchmark. All rows use 96 shared QUBO-training trajectory evaluations and 24 true post-QUBO candidate evaluations per seed.

method	runs	refinement_success_rate	refined_nominal_error_median	refined_selected_worst_error_median	refined_nominal_fuel_median	solver_true_evaluations_median	shared_qubo_training_evaluations_median	runtime_seconds_median
surrogate_qubo_sa	30	0.9333	0.0690	0.0941	0.0917	24.0000	96.0000	0.2952
exact_qubo_top24	30	0.9333	0.0690	0.0941	0.0917	24.0000	96.0000	0.2967
qaoa_random_p1	30	0.5000	0.0800	0.1055	0.0875	24.0000	96.0000	0.3222
qaoa_optimized_p1	30	0.7000	0.0682	0.0933	0.0917	24.0000	96.0000	0.3212
qaoa_optimized_p2	30	0.9667	0.0690	0.0941	0.0917	24.0000	96.0000	0.3491

Table 42: Legacy 10-seed QAOA-depth ablation retained for provenance. The manuscript's QAOA-depth statistical result uses the 30-seed package.

method	runs	refinement_success_rate	refined_nominal_error_median	refined_selected_worst_error_median	refined_nominal_fuel_median	solver_true_evaluations_median	shared_qubo_training_evaluations_median	runtime_seconds_median
surrogate_qubo_sa	10	1.0000	0.0682	0.0933	0.0917	24.0000	96.0000	0.3912
exact_qubo_top24	10	0.9000	0.0690	0.0941	0.0917	24.0000	96.0000	0.3957
qaoa_random_p1	10	0.3000	0.0948	0.1203	0.0833	24.0000	96.0000	0.4140
qaoa_optimized_p1	10	0.6000	0.0690	0.0941	0.0917	24.0000	96.0000	0.4098
qaoa_optimized_p2	10	0.9000	0.0690	0.0941	0.0917	24.0000	96.0000	0.4964

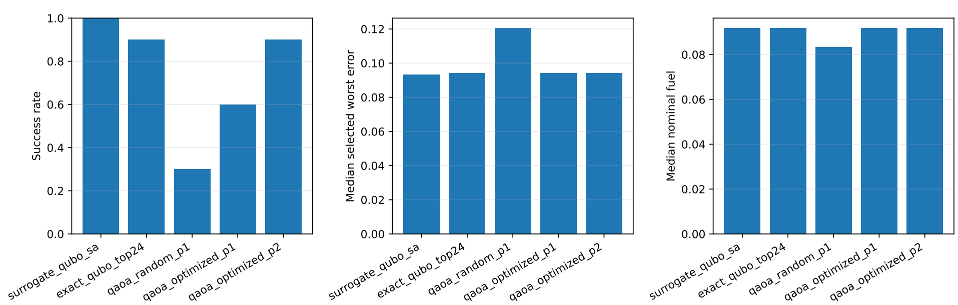

Figure 21: Legacy 10-seed QAOA-depth summary retained as provenance.

9 Teacher-Controlled Diagnostics

The teacher-controlled benchmark is feasible by construction and should not be confused with an operational transfer. It is retained because it diagnoses whether the architecture can recover a known feasible low-thrust target and because the oracle continuous-control diagnostic shows that continuous initialization can dominate the outcome. The teacher-controlled tables are small-seed evidence and do not support a statistically significant method ranking.

Table 43: Teacher-controlled benchmark results. Values are medians across three seeds. QUBO and QAOA totals include the shared QUBO-training trajectory evaluations.

Benchmark	Method	Comparison group	Refined nominal error	Selected recovery worst error	All-mask diagnostic worst error	Active fraction	Solver true evals	Shared QUBO train evals	Total true evals	Refine success
teacher_controlled	random	method_comparison	0.0115	0.0137	1.1009	0.3750	52.0000	0.0000	52.0000	0.6667
teacher_controlled	cross_entropy	method_comparison	0.0381	0.2626	0.7695	0.3750	56.0000	0.0000	56.0000	0.3333
teacher_controlled	genetic	method_comparison	0.0213	0.5602	0.8393	0.3125	52.0000	0.0000	52.0000	0.0000
teacher_controlled	true_sa	method_comparison	0.6349	0.9638	16.4017	0.6250	52.0000	0.0000	52.0000	0.0000
teacher_controlled	surrogate_qubo_sa	method_comparison	0.0263	0.2162	0.8145	0.3750	20.0000	32.0000	52.0000	0.3333
teacher_controlled	qaoa_statevector	method_comparison	0.2495	0.2496	1.0213	0.3125	20.0000	32.0000	52.0000	0.3333
teacher_controlled	all_windows_continuous	method_comparison	6.3495	8.4578	17.8387	1.0000	3.0000	0.0000	3.0000	0.0000
teacher_controlled	teacher_seeded_schedule	method_comparison	1.2183	0.9547	1.7652	0.4375	30.0000	0.0000	30.0000	0.0000
teacher_controlled	teacher_controls_oracle_diagnostic	oracle_diagnostic	0.0067	0.0080	1.1910	0.4375	7.0000	0.0000	7.0000	1.0000

Table 44: Teacher-controlled recovery fuel and refinement effort. Fuel is reported in normalized acceleration-time units.

Method	Nominal fuel	Mean recovery fuel	Max recovery fuel	Refinement nfev	Recovery success
random	0.0579	0.0579	0.0579	24.0000	0.6667
cross_entropy	0.0892	0.0813	0.0813	30.0000	0.3333
genetic	0.0679	0.0650	0.0650	30.0000	0.0000
true_sa	0.1592	0.1355	0.1438	16.0000	0.0000
surrogate_qubo_sa	0.0733	0.0650	0.0650	27.0000	0.3333
qaoa_statevector	0.0488	0.0488	0.0488	30.0000	0.3333
all_windows_continuous	0.2595	0.2194	0.2284	3.0000	0.0000
teacher_seeded_schedule	0.1134	0.0793	0.0849	30.0000	0.0000
teacher_controls_oracle_diagnostic	0.0281	0.0272	0.0283	7.0000	1.0000

Table 45: Teacher-controlled QUBO surrogate fit diagnostics.

seed	ridge	train_rmse	val_rmse	train_r2	val_r2	val_spearman	val_pairwise_order_accuracy	val_top_k	val_top_k_recall	n_train	n_val	shared_qubo_training_evaluations	qubo_training_true_evaluations	equal_total_budget
11	0.0001	0.0001	9.4732	1.0000	0.0443	0.7143	0.7857	2	0.5000	24	8	32	32	True
23	0.0001	0.0002	13.1807	1.0000	-1.7931	0.6190	0.7857	2	1.0000	24	8	32	32	True
37	0.0001	0.0001	7.2534	1.0000	0.6466	0.5714	0.6786	2	1.0000	24	8	32	32	True

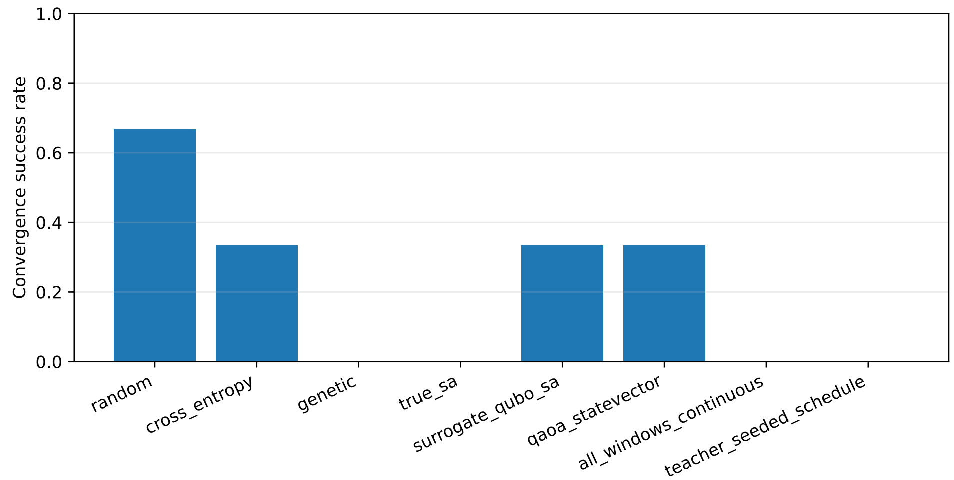

Figure 22: Teacher-controlled selected-branch refinement success rates. The result demonstrates nonzero recovery success but not quantum advantage.

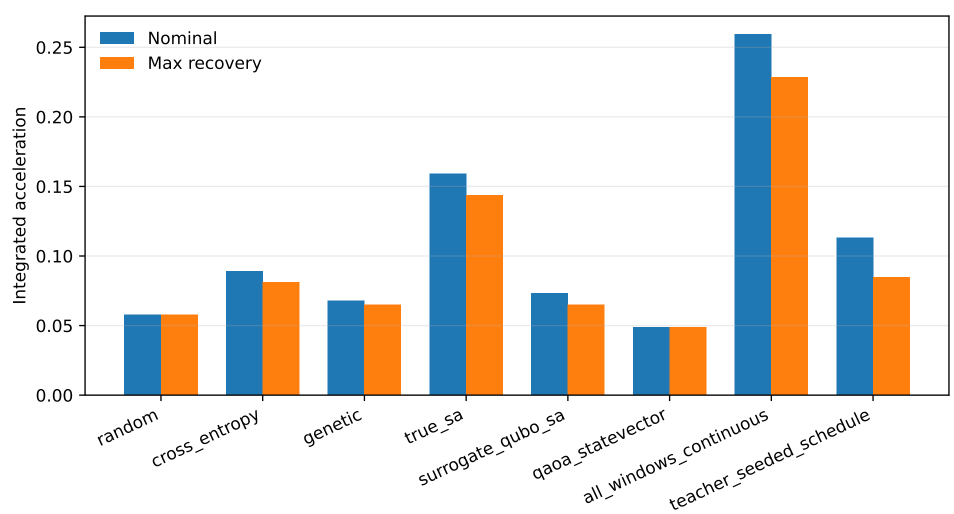

Figure 23: Teacher-controlled nominal and recovery fuel summaries. Fuel differences must be interpreted together with feasibility and missed-thrust error.

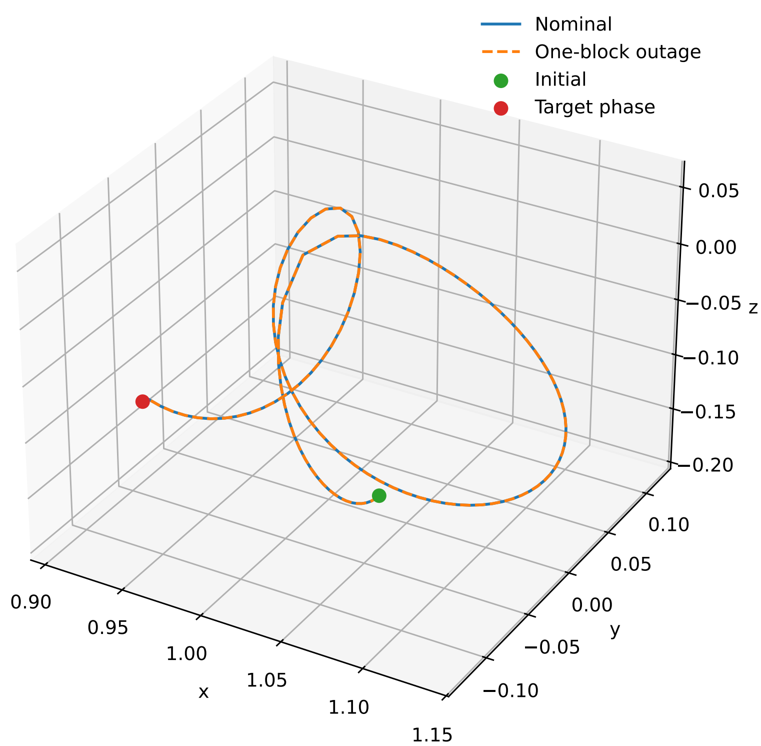

Figure 24: Example trajectory from the teacher-controlled benchmark. The case is a normalized CR3BP algorithm benchmark, not a flight-ready trajectory.

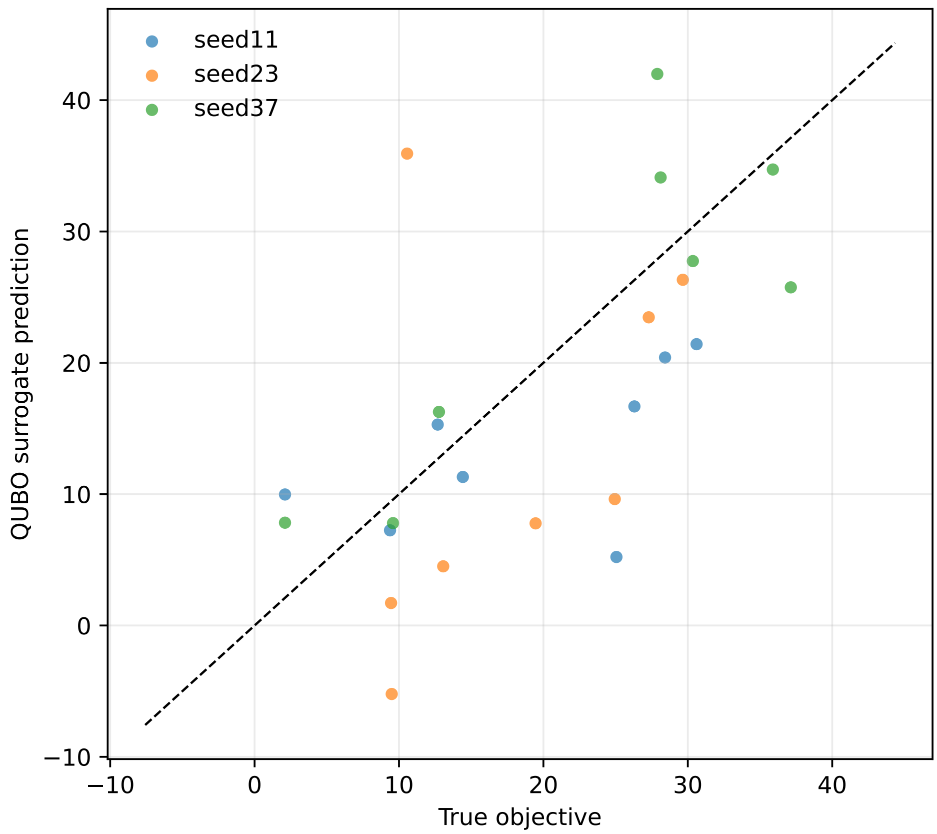

Figure 25: Teacher-controlled QUBO fit diagnostic. Surrogate quality remains mixed across the small teacher-controlled seeds.

10 Artifact and Provenance Notes

The machine-readable manifest ../data/results/artifact_manifest.json records scoped SHA-256 hashes and byte counts for source, configuration, manuscript, result, table, cache, and figure artifacts. The hashes and byte counts are authoritative for artifact identity. The threshold-sensitivity artifacts in ../data/results/phase_shift_cardinality_30seed/threshold_sensitivity.csv and ../tables/phase_shift_cardinality_30seed/threshold_sensitivity_table.tex are derived directly from the recorded 30-seed raw CSV without rerunning trajectory optimization. The claim-evidence ledger artifacts in ../data/results/claim_evidence_ledger/ and ../tables/claim_evidence_ledger/ separate selected-branch evidence, all-mask diagnostics, all-configured-mask evidence, the new independent-HS all-configured headroom row, the independent-HS branch replay row, the positive independent-HS simple bicircular phase-stress row, the positive independent-HS cached-Horizons-derived solar-tidal replay row, the independent-HS cached-Horizons point-mass retuning row, the independent-HS multi-epoch cached-Horizons point-mass retuning row, the independent-HS SPICE-derived ephemeris replay row, the completed accepted-control replay row, the cached Horizons force-model contrast row, the negative hard-catalog bicircular solar-tidal stress row, and the completed negative bicircular retuned recovery row. The independent-HS all-configured headroom artifacts in ../data/results/independent_hs_all_configured_headroom/, ../tables/independent_hs_all_configured_headroom/, and ../figures/independent_hs_all_configured_headroom/ optimize all eight configured one-segment masks for the p=0.4, $a_{\max} = 0.2$ phase-shift row under normalized CR3BP; they are not high-fidelity validation, broad production-solver validation/parity, or broader outage-family robustness evidence. The independent-HS bicircular phase-stress artifacts in ../data/results/independent_hs_bicircular_phase_stress/ and ../tables/independent_hs_bicircular_phase_stress/independent_hs_bicircular_phase_stress_table.tex replay the converged polish row under a simple circular solar-tidal stress model and report a positive threshold result; they are not SPICE/high-fidelity validation or broad production-solver validation/parity. The independent-HS cached-Horizons-derived solar-tidal replay artifacts in ../data/results/independent_hs_horizons_solar_tidal_replay/, ../tables/independent_hs_horizons_solar_tidal_replay/independent_hs_horizons_solar_tidal_replay_table.tex, and ../data/cache/horizons/independent_hs_phase_shift_2026jan01_vectors.json replay the same persisted controls with representative 2026-Jan-01 cached JPL Horizons geometry; they are not SPICE/high-fidelity/flight validation or broad production-solver validation/parity. The independent-HS cached-Horizons point-mass retuning artifacts in ../data/results/independent_hs_horizons_point_mass_retuning/, ../tables/independent_hs_horizons_point_mass_retuning/independent_hs_horizons_point_mass_retuning_table.tex, and ../data/cache/horizons/independent_hs_phase_shift_2026jan01_vectors.json document failed direct replay followed by independent least-squares retuning under a geocentric Earth/Moon/Sun point-mass model; they are not SPICE/full high-fidelity/flight validation, broad production-solver validation/parity, fuel optimality, DOI evidence, quantum, QUBO, or QAOA evidence. The independent-HS multi-epoch point-mass retuning artifacts in ../data/results/independent_hs_horizons_multi_epoch_point_mass_retuning/, ../tables/independent_hs_horizons_multi_epoch_point_mass_retuning/independent_hs_horizons_multi_epoch_point_mass_retuning_table.tex, and the committed ../data/cache/horizons/independent_hs_phase_shift_2026*_vectors.json caches extend that stress/retuning check across four fixed 2026 epochs, but remain point-mass retuning artifacts rather than SPICE/full high-fidelity/flight validation, broad production-solver validation/parity, fuel optimality, DOI evidence, quantum, QUBO, or QAOA evidence.

The independent-HS SPICE-derived replay artifacts in ../data/results/independent_hs_spice_ephemeris_replay/, ../tables/independent_hs_spice_ephemeris_replay/independent_hs_spice_ephemeris_replay_table.tex, and the committed compact caches ../data/cache/spice/independent_hs_phase_shift_2026*_spice_vectors.json replay the same 36 already-retuned controls without rerunning optimization; all rows pass, worst nominal/branch replay errors are 0.021439441253166033/0.024730650824609506, and maximum Horizons-to-SPICE replay delta is $3.2763991519857427 \times 10^{-10}$. They are point-mass ephemeris-source replay artifacts rather than full high-fidelity/flight validation, broad production-solver validation/parity, fuel optimality, DOI evidence, quantum, QUBO, or QAOA evidence. The historical tail-coast branch audit summarizes recorded JSON only; the focused branch-control replay artifacts in ../data/results/hard_catalog_tail_coast_branch_control_replay/ and ../tables/hard_catalog_tail_coast_branch_control_replay/ repropagate persisted accepted controls under normalized CR3BP without rerunning optimization or claiming high-fidelity validation. The Horizons contrast artifacts in ../data/results/horizons_ephemeris_force_model_contrast/ and ../tables/horizons_ephemeris_force_model_contrast/horizons_ephemeris_force_model_contrast_table.tex, backed by the cache in ../data/cache/horizons/, compare simple bicircular assumptions against cached ephemeris geometry only; they are not SPICE validation or accepted-control high-fidelity replay. The hard-catalog bicircular stress artifacts in ../data/results/bicircular_solar_tidal_stress/ and ../tables/bicircular_solar_tidal_stress/bicircular_solar_tidal_stress_table.tex repropagate the same controls under a simple circular solar tidal term and report threshold failures; they are not SPICE ephemeris validation or broad production-solver validation/parity. The bicircular retuned recovery artifacts in ../data/results/bicircular_tail_coast_recovery/ and ../tables/bicircular_tail_coast_recovery/bicircular_tail_coast_recovery_table.tex retune all 27 configured one/two-segment masks at fixed Sun phase 0° but remain a negative threshold result; they are not SPICE/high-fidelity/flight validation, broad production-solver validation/parity, fuel optimality, quantum, QUBO, or QAOA evidence. The replay/stress validation artifacts in ../data/results/replay_stress_validation/ and ../tables/replay_stress_validation/replay_stress_validation_table.tex repropagate recorded nominal-control sidecars only; they do not rerun least-squares optimization, replay branch recovery controls, or claim high-fidelity validation. The manifest intentionally excludes itself, virtual environments, runtime caches, logs, LaTeX auxiliary files, and raw NAIF kernel binaries; the committed Horizons and compact SPICE-derived vector caches are included as data artifacts. Its `git_head_at_generation` field records the repository state when the manifest was generated; a committed manifest necessarily records the parent/pre-final state rather than the final commit containing the refreshed manifest. Historical experiment metadata may predate the repository or contain dirty-state records, so current submission provenance is verified by clean git status after final commit, manifest checking, tests, and local LaTeX builds.